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1.0 INTRODUCTION

1.1 PROJECT BACKGROUND

Diabetes is a chronic health condition caused by the body's failure to effectively regulate blood glucose levels, either due to insufficient insulin production or the inability to use insulin properly. If left unmanaged, diabetes may lead to serious complications such as hypertension, heart disease and other long term health problems. In the United States, diabetes remains a significant public health concern affecting individuals across various age groups, genders and races. As a result, analysing diabetes related data is important for understanding population health trends and identifying significant risk factors for the disease.

This project is based on a health dataset that contains patient level records related to diabetes cases in the United States. The dataset includes demographic variables such as age, age group, gender, race, year and location along with clinical and lifestyles attributes including Body Mass Index (BMI), BMI category, blood glucose level, HbA1c level, hypertension status, heart disease status, smoking history and diabetes status. These variables give a complete picture of both demographic characteristics and health risk indicators. Using this data, interactive dashboards are created to explore patterns, compare diabetic and non-diabetic patients and assess how derived variables such as age group, BMI category and race contribute to a better understanding of diabetes risk patterns.

The dashboards created in this project are intended to help public health analysts, healthcare practitioners and medical researchers analyse diabetes related trends and risk factors. By presenting the findings in clear and interactive visualisations, the initiatives promote effective data exploration and data driven insights important to health analysis and research.

1.2 PROBLEM STATEMENT

The complexity and variability of clinical data make it difficult for healthcare providers to identify people at high risk of diabetes, even with the availability of detailed patient information. Analysis is made more difficult by the mixed data types, which makes it possible for any class imbalance and diverse measurement scales. The dataset comprises 100,000 patient records with demographic, lifestyle and biomedical information. Strong preprocessing and analytical methods are needed to extract significant patterns from such diverse data.

Additionally, it is challenging to manually interpret relationships between predictors and diabetes outcomes due to the enormous amount of data. These difficulties highlight the necessity of systematic data analysis to identify important contributing factors and produce trustworthy insights that can enhance screening effectiveness, promote early detection and direct preventive healthcare initiatives.

1.3 DATA CONTEXT

The dataset utilized for this project is the Diabetes Clinical Dataset, which was deliberately chosen from Kaggle open-source repository which is accessible at <https://www.kaggle.com/datasets/ziya07/diabetes-clinical-dataset100k-rows/data>. This dataset acts as a specialised clinical repository for monitoring and evaluating individuals' health profiles in relation to their diabetes diagnosis. It offers a multidimensional view of health, combining demographic information with important physiological parameters to allow for a data-driven understanding of the disease's risk factors.

In terms of data scope, the collection represents a longitudinal perspective, gathering clinical information over eight-year period from 2015 to 2022. This timeframe is critical for tracking health trends and the evolution of patient situations in the modern era. Geographically, the dataset is diverse, containing information from 55 distinct locations across multiple states and territories in the United States, including Alabama, Alaska, Arizona, Arkansas and California. This wide geographic coverage supports a more comprehensive representation of patient records across different regions rather than being confined to a single location. With a total of 100,000 record of patients, the dataset provides a substantial sample size that supports reliable descriptive and comparative analysis across different patient groups.

The structural context of the data is defined by 17 major clinical and demographic characteristics. These contain basic patient identifiers and demographics such as year, age, location, race which help identify high-risk patients. In addition, the data includes crucial medical history indicators, notably hypertension, heart disease and smoking history, all of which are identifies as diabetes co-morbidities. Most significantly, the dataset includes objective clinical measure such as BMI, HbA1c and blood glucose levels. The last goal variable, Status of Diabetes, classifies patients as diabetic or non-diabetic giving the necessary ground truth for establishing correlations between the mentioned health indicators and the final clinical diagnosis.

1.4 TARGET AUDIENCE

The primary target audience for this project includes public health analysts, healthcare practitioners, and medical researchers who play a critical role in analysing patient-level records related to diabetes cases in the United States. Public health analysts and rely on accurate, data-driven insights to understand population-level diabetes trends, risk factors, and health disparities. Healthcare practitioners can use these insights to support clinical decision-making and identify patient groups at higher risk. Medical researchers benefit from comprehensive visualizations that enable deeper exploration of demographic, biological, and lifestyle factors associated with diabetes. By developing an interactive dashboard, this project will help these professionals efficiently interpret large healthcare datasets, identify high-risk populations, and support evidence-based strategies for diabetes prevention, management, and policy development.

2.0 OBJECTIVES

2.1 PROJECT AIMS

The aim of this project is to perform a thorough analysis of a large-scale diabetes clinical dataset in order to identify significant associations between the presence of diabetes and biological, lifestyle and demographic factors. The project aims to improve knowledge of how different health indicators contribute to diabetes risk in various populations by investigating these variables.

By analyzing the differences in biological, lifestyle and demographic factors between people with and without diabetes, this project seeks to understand the roots of the disease. This project also aims to convert complex clinical data into valuable insights by gradually analyzing population profiles, comparing various health risk factors and identifying deeper patterns through derived health indicators. The results are intended to improve knowledge of diabetes risk patterns and facilitate better decision-making in healthcare screening, prevention and policy planning through a dashboard-driven analytical approach.

2.2 RESEARCH QUESTIONS

The development of research questions is an important stage in establishing a clear direction for the data analysis process. These questions serve as a link between the overarching project goals and the specific insights gleaned from the Diabetes Clinical Dataset; by addressing these issues, the study aims to find underlying trends and connections in the 100,000 patient records from 2015 to 2022. The research topics are classified based on respective Research Objectives, as follows:

The first set of questions focusses on the demographic and health distribution patterns among patients:

- Research Question 1.1: What is the distribution and trend of diabetes status from 2015 to 2022?
- Research Question 1.2: How are diabetic and non-diabetic patients distributed by gender, age group and race?
- Research Question 1.3: Does smoking history differ by gender?
- Research Question 1.4: How are diabetes cases geographically distributed across US states?

- Research Question 1.5: What is the proportion of patients with hypertension or heart disease among age groups?

The second set of questions explores to compare health risk indicators between diabetic and non-diabetic patients:

- Research Question 2.1: How does average BMI differ between diabetic and non-diabetic patients across BMI categories?
- Research Question 2.2: How do HbA1c level and blood glucose level differ between diabetic and non-diabetic patients?
- Research Question 2.3: How does the prevalence of hypertension differ between diabetic and non-diabetic patients?
- Research Question 2.4: How does the prevalence of heart disease differ between diabetic and non-diabetic patients?
- Research Question 2.5: How does smoking history differ between diabetic and non-diabetic patients?

The third set of questions explains on how to examine the distribution and trends of diabetes prevalence:

- Research Question 3.1: How does diabetes prevalence vary by age group across gender?
- Research Question 3.2: How has the distribution of diabetes cases across BMI categories changed from 2015 to 2022?
- Research Question 3.3: How has the number of diabetes cases changed from 2015 to 2022 across gender categories?

2.3 RESEARCH OBJECTIVES

The main objective of this project is to analyse diabetes cases in the United States by examining demographic characteristics, health indicators and derived health variables using data visualization techniques. To achieve this aim, the specific research objectives of this study are as follows:

1. To identify demographic and health distribution patterns among patients.
2. To compare health risk indicators between diabetic and non-diabetic patients.
3. To examine the distribution and trends of diabetes prevalence.

3.0 METHODOLOGY

3.1 GENERIC DATA SCIENCE LIFE CYCLE

This project adopts a Generic Data Science Life Cycle to provide a structured framework for conducting the analysis, from data acquisition to the interpretation of results. The life cycle consists of five main phases, namely Obtain, Scrub, Explore, Model, and Interpret, which guide the systematic transformation of raw data into meaningful insights. Figure 3.1 illustrates the Generic Data Science Life Cycle used in this study and serves as the conceptual foundation for the methodology adopted in the subsequent sections.

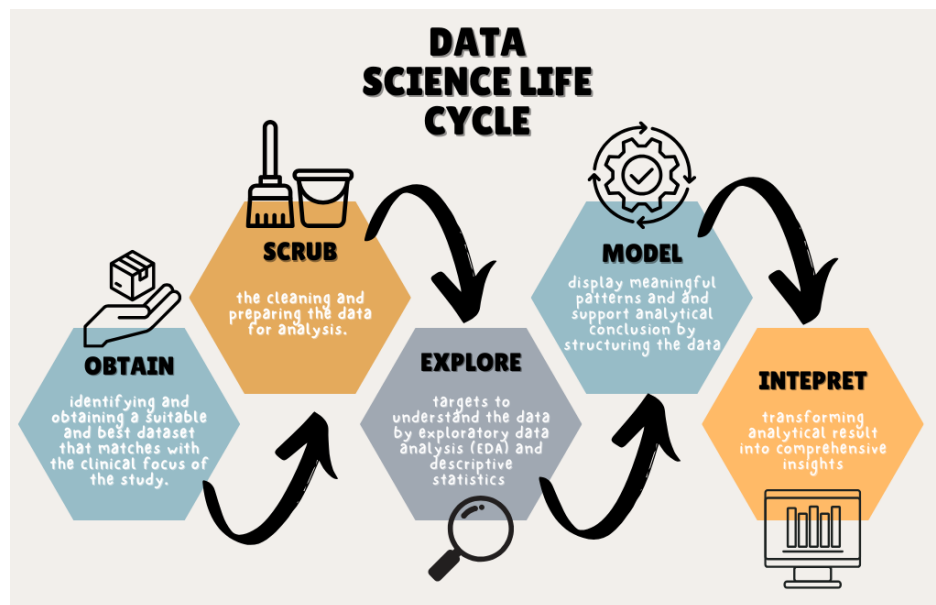


Figure 3.1 Generic Data Science Life Cycle

3.2 STAGES AND ACTIVITIES

The methodology for this research study is based on the Data Science Life Cycle, which ensures a systematic transition from raw data collection to relevant clinical insights. As illustrated in the life cycle framework, the analysis follows five main phases which are Obtain, Scrub, Explore, Model and Interpret. The Obtain phase began with the retrieval of the Diabetes Clinical Dataset, which included 100,000 data form the Kaggle repository to serve as the study's foundation. This was followed by the Obtain phase, which included many data cleansing procedures to assure excellent data quality. This stage's key actions included duplication elimination, missing value management and variable transformation into appropriate data types. Furthermore, this phase includes the creation of derivative variables such as BMI Category and Age Group allow for more detailed demographic research.

After refining the data, the project went on to the Explore phase, which involved Exploratory Data Analysis (EDA) to find initial patterns and statistical distributions. During this stage, descriptive statistics were used to summarise the clinical metrics of 100,000 patients with an emphasis on the distribution of blood glucose and HbA1C levels across different locations and genders. These exploratory discoveries influenced the Model Phase, in which the analytical focus moved to answering particular research objectives. This included conducting comparative assessments of diabetic and non-diabetic patients and investigating the relationships between critical health indicators such as BMI and hypertension status.

The fifth element of the process was the Interpretation phase, which focused on communicating findings through visualizations. The major task was to design and construct an interactive Power BI Dashboard that combined clinical measures and demographic data into a user-friendly interface. This phase finished with the production of a compelling narration that matched the visual outputs to the defined Research Objectives, ensuring that the final insights into diabetes risk factors were clear, evidence-based, and accessible to the target audience.

3.3 GANTT CHART AND MILESTONES

Figure 3.2 presents the Gantt chart outlining the project timeline and key milestones across each phase of the data science life cycle.

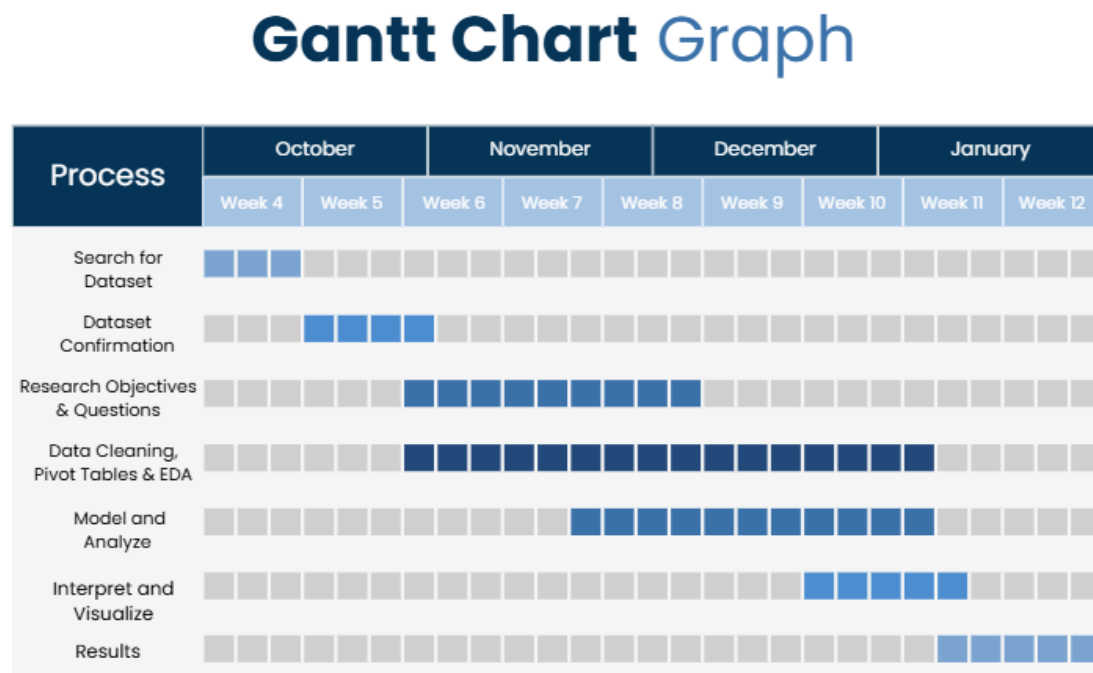


Figure 3.2 Gantt Chart and Milestones

3.4 DATA DESCRIPTION

The Table 3.1 shows data description for each attribute as shown as below:

Table 3.1 Data Description

No	Variable Name	Descriptions
1	Year	The year in which the record was collected from 2015 - 2022.
2	Gender	The biological sex of the individual with 3 categories which Female, Male and Other
3	Age	Actual age in years.
4	Location	The geographic area or region of the individual of 55 areas in United States.
5	Hypertension	Indicates if the individual has hypertension.
6	Heart Disease	Indicates if the individual has heart disease.
7	Smoking History	Smoking status that is divided into 6 categories which are Current, Ever, Former, Never, No Info and Not Current.
8	BMI	Body Mass Index value.
9	HbA1c Level	Haemoglobin A1c level (%) which is a measurement of long- term blood glucose control.
10	Blood Glucose Level	Measured blood glucose concentration.
11	Status of Diabetes	Indicates diabetes status which is yes or no.
12	Age Group	Grouped age range categories which are Adult, Child, Senior, Teenager and Young Adult.
13	Race	The race or ethnicity of the individual which are African American, Asian, Other, Hispanic and Caucasian.
14	BMI Category	Classification of BMI which are Normal, Obese, Overweight and Underweight.

3.5 DATA DICTIONARY

The Table 3.2 shows data dictionary for each attribute as shown as below:

Table 3.2 Data Dictionary

No	Variable Name	Variable Description	Data Types	Level of Measurement
1	Year	The year in which the record was collected.	Integer	Interval
2	Gender	The biological sex of the individual	Categorical	Nominal
3	Age	Actual age in years.	Integer	Ratio
4	Location	The geographic area or region of the individual.	Categorical	Nominal
5	Hypertension	Indicates if the individual has hypertension	Binary	Nominal
6	Heart Disease	Indicates if the individual has heart disease	Binary	Nominal
7	Smoking History	Smoking status	Categorical	Nominal
8	BMI	Body Mass Index value	Float	Ratio
9	HbA1c Level	Haemoglobin A1c level (%) which is a measurement of long-term blood glucose control	Float	Ratio
10	Blood Glucose Level	Measured blood glucose concentration	Integer	Ratio
11	Status of Diabetes	Indicates diabetes status	Binary	Nominal
12	Age Group	Grouped age range categories	Categorical	Ordinal
13	Race	The race or ethnicity of the individual.	Categorical	Nominal
14	BMI Category	Classification of BMI	Categorical	Ordinal

3.6 EXPLORATORY DATA ANALYSIS

3.6.1 OBJECTIVE 1

1) What is the distribution and trend of diabetes status from 2015 to 2022?

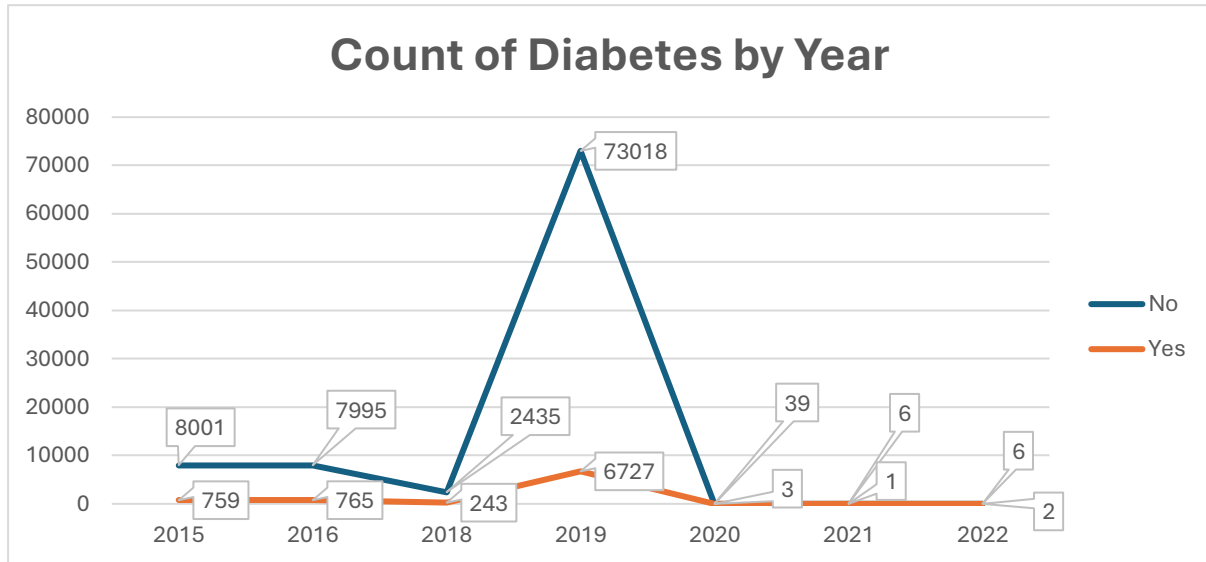


Figure 3.3 Count of Diabetes by Year

Figure 3.3 indicates that the distribution and trend of diabetes status for non-diabetic patients continuously outnumbering diabetic patients in every year. From 2015 and 2016, there were about 8,000 non-diabetic patients, whereas there were less than 800 diabetes cases. With non-diabetic cases peaking at 73,018 and diabetes cases increased to 6,727 in 2019, both groups exhibit a fall in 2018 and a dramatic spike in 2019, indicating a significant growth in data coverage for that year. Following 2019, there is a major drop in both diabetic and non-diabetic populations from 2020 to 2022, with extremely low levels noted.

2) How are diabetic and non-diabetic patients distributed by gender, age group and race?

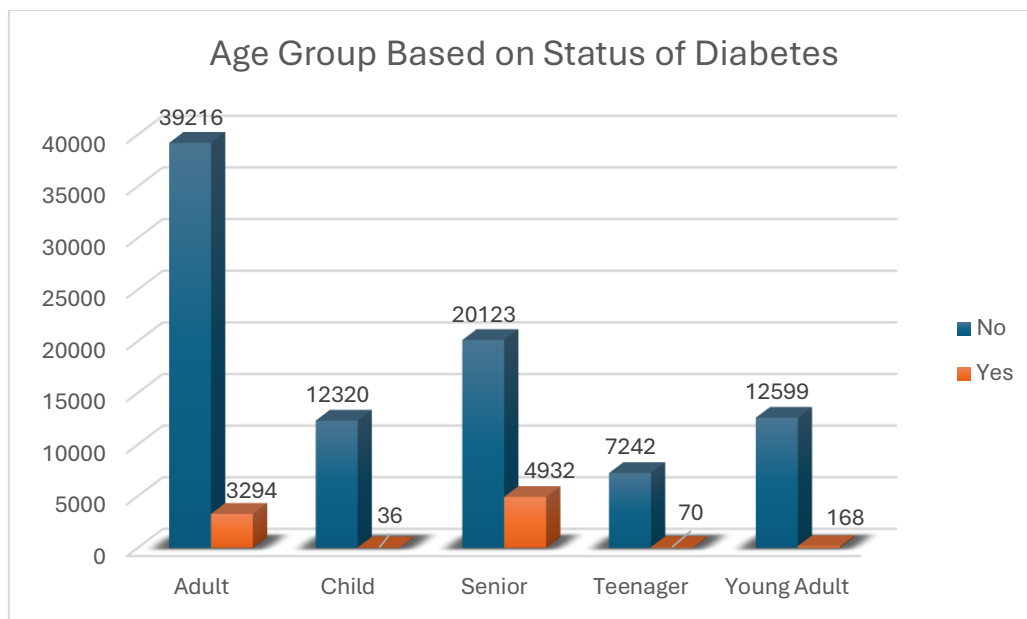


Figure 3.4 Age Group Based on Status of Diabetes

Figure 3.4 shows the distribution of diabetic and non-diabetic patients among age categories. Overall, adults had the most patients with 39,216 non-diabetic and 3,294 diabetic cases, demonstrating that diabetes is more common among adults than in younger categories. This is followed by the senior category, which includes 20,123 non-diabetic and 4,932 diabetic patients, indicating a rather high prevalence of diabetes among the elderly. In contrast, the child and teenager categories had very low numbers of diabetic patients which is 36 and 70 cases respectively, despite having a considerable percentage of non-diabetics. The young adult population also had a low prevalence of diabetes, with 168 diabetics compared to 12,599 non-diabetics.

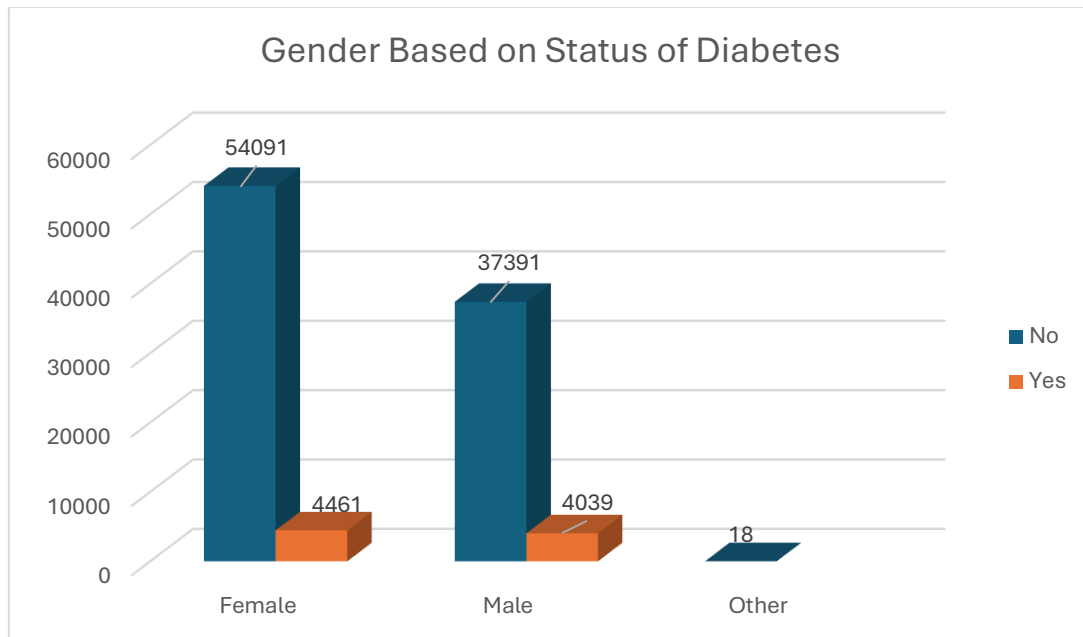


Figure 3.5 Gender Based on Status of Diabetes

Figure 3.5 illustrates the distribution of diabetic and non-diabetic patients based on gender. According to the findings, female patients form the largest group of non-diabetic individuals, with 54,091 and 4,461 cases respectively. Similarly, male patients reported 37,391 non-diabetic and 4,039 diabetic instances, indicating a roughly equal proportion of diabetic cases between males and females, while females had a slightly larger count. The number of patients classified as other is small, with only 18 non-diabetic cases and no recorded diabetic cases. Overall, while females make up a bigger proportion of the population, the prevalence of diabetes appears to be relatively similar across both genders, implying that gender may not be a powerful predictor of diabetes status in this study.

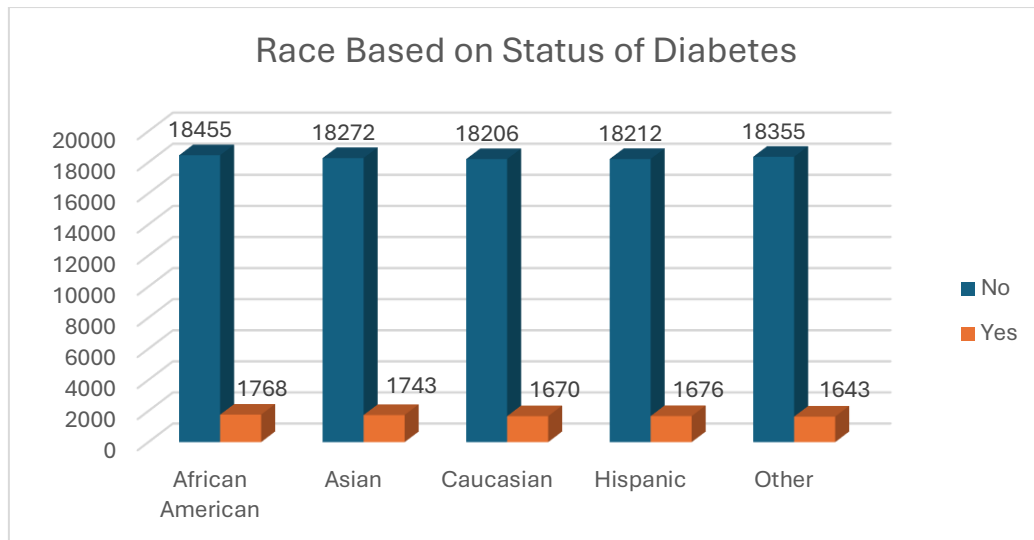


Figure 3.6 Race Based on Status of Diabetes

Figure 3.6 illustrates the distribution of diabetic and non-diabetic patients across different racial groups. Across all races, non-diabetic individuals form the majority, with counts remaining relatively consistent at around 18,000 per group. The number of diabetic cases varies from roughly 1,543 to 1,768 across racial categories, which is significantly lower yet reasonably equally distributed. The “Other” category has the lowest number of diabetic patients, while the African American and Asian groupings have slightly higher percentages than the other racial categories.

3) Does smoking history differ by gender?

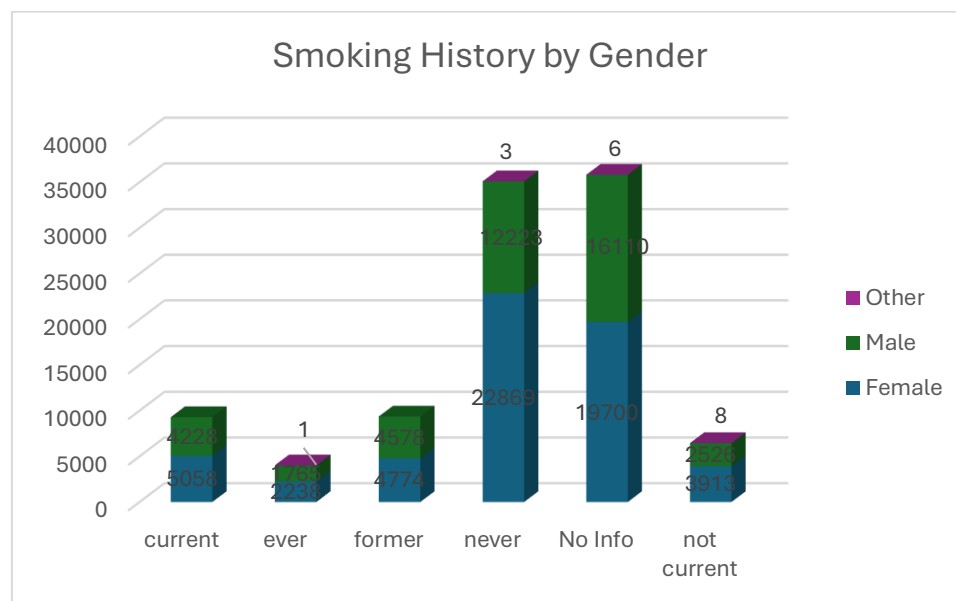


Figure 3.7 Smoking History by Gender

Based on Figure 3.7, the distribution of smoking history by gender shows that females dominate the majority of identified nonsmokers, with 22,869 females reporting never smoking compared with 12,223 males. Females (5,058) outweigh males (4,228) among current smokers, whereas former smokers (4,774 females and 4,578 males) are virtually evenly split. However, one important disadvantage of this dataset is the large amount of missing or hidden information, as the “No Info” group includes over 35,000 patients of both genders.

4) What is the proportion of patients with heart disease or hypertension among age groups?

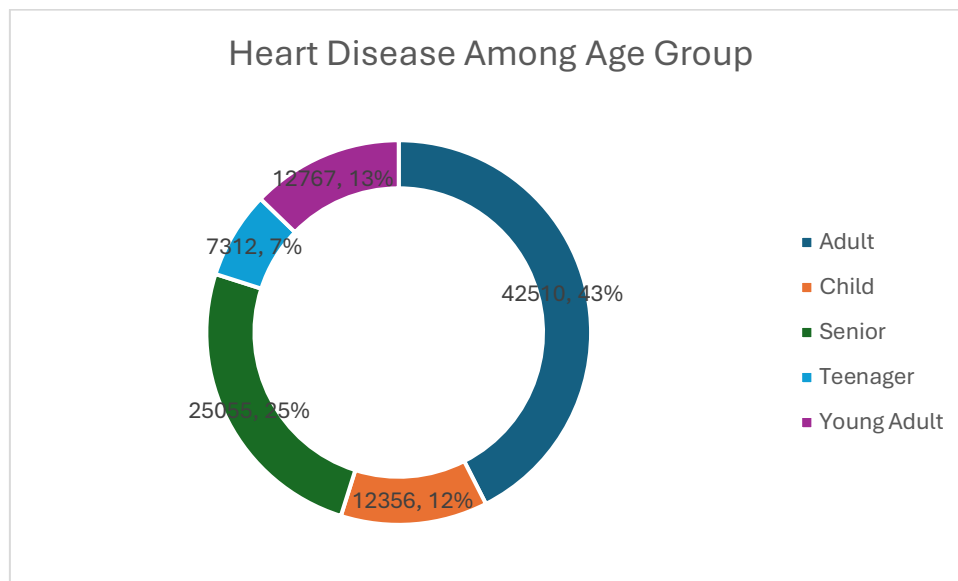


Figure 3.8 Heart Disease by Age Group

According to Figure 3.8, adults represent for the largest share of heart disease cases at approximately 43% with 42,510 cases followed by seniors at 25% with 25,055 cases. Younger age groups contribute smaller proportions, with young adults representing 13% with 12,767 cases, children for 12% with 12,356 cases and teenagers the lowest at 7% with 7,312 cases. Overall, the distribution indicates that heart disease cases are mainly focused among adult and senior age groups, while younger populations represent a relatively smaller proportion.

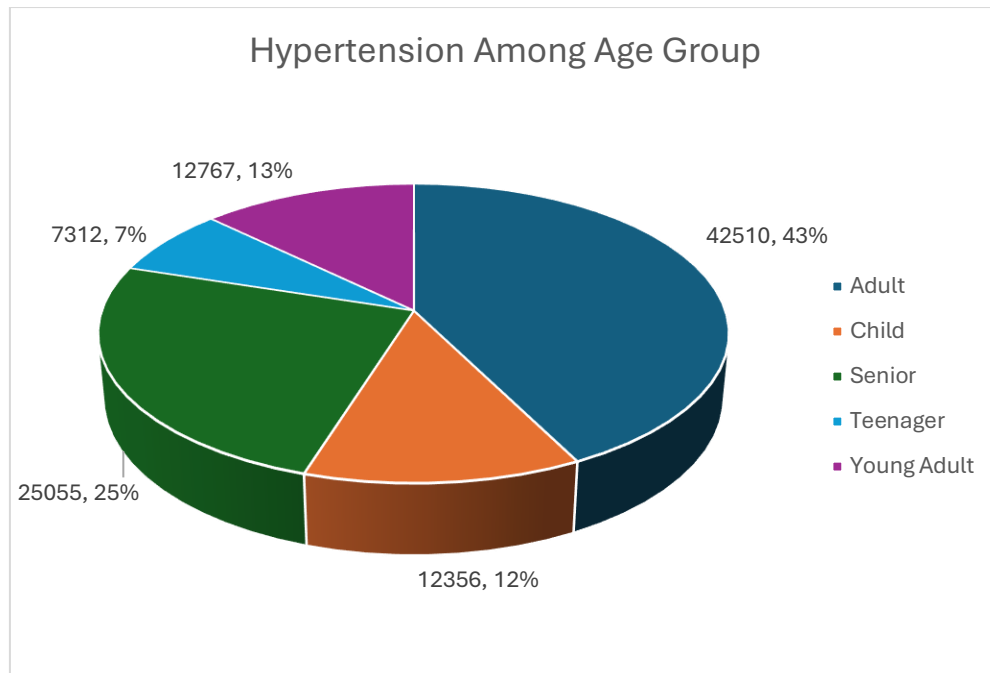


Figure 3.9 Hypertension Among Age Groups

Based on Figure 3.9 adults represent for the largest share of hypertension cases at approximately 43% with 42,510 cases followed by seniors at 25% with 25,055 cases. Younger age groups contribute smaller proportions, with young adults representing 13% with 12,767 cases, children for 12% with 12,356 cases and teenagers the lowest at 7% with 7,312 cases. Overall, the distribution indicates that hypertension cases are mostly concentrated among adult and senior age groups, while younger populations represent a relatively smaller proportion.

3.6.2 OBJECTIVE 2

- 1) How do average BMI and BMI categories differ between diabetic and non-diabetic patients?

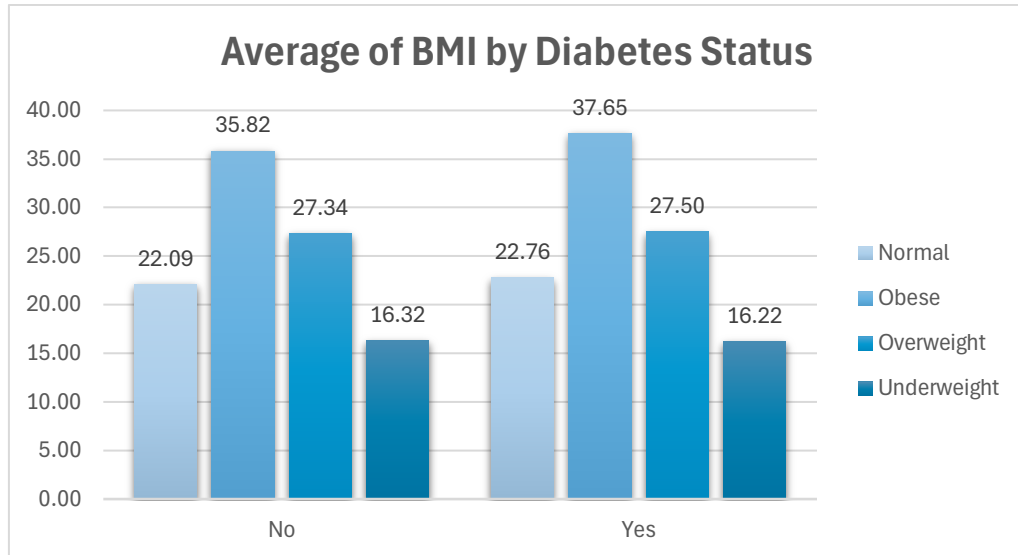


Figure 3.10 Average of BMI by Diabetes Status

The chart above shows the average of BMI by diabetes status among the patients. Patients with diabetes consistently have slightly higher average BMI values than the one without diabetes across all BMI categories excluding Underweight patients. The greatest difference can be seen in the Obese category, where the average BMI of diabetics is 37.65, significantly higher than the non-diabetics (35.82). This implies a strong correlation between the risk of diabetes and a higher BMI. Diabetics also have slightly higher BMI averages even within Overweight (27.34 and 27.5) and Normal (22.09 and 22.76) categories. This suggests that while BMI may not fully capture the complexity of diabetes risk, it is still a significant indicator, particularly in the obese patients.

- 2) How do HbA1c level and blood glucose level differ between diabetic and non-diabetic patients?

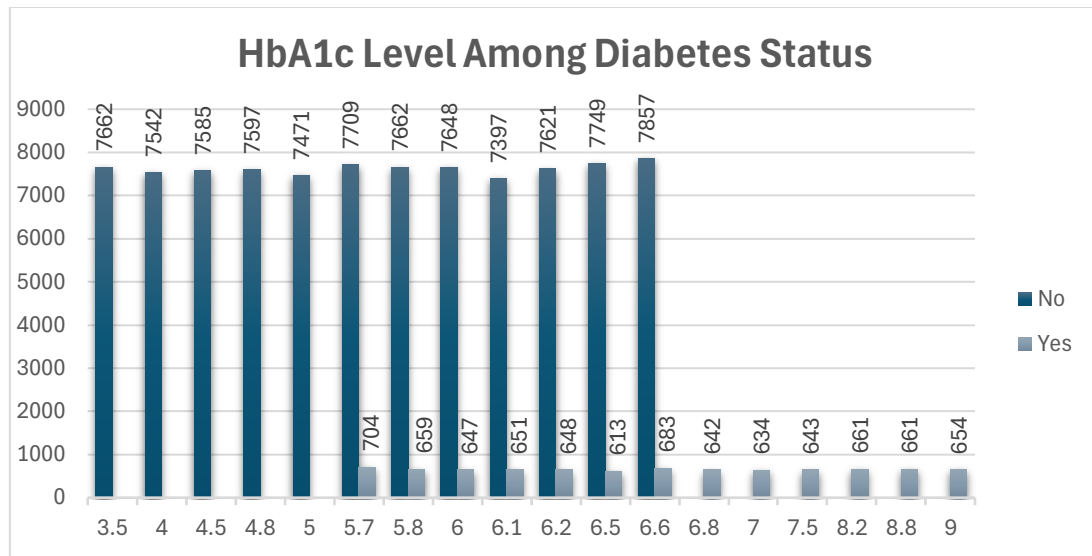


Figure 3.11 BMI Category Distribution by Diabetes Status

The chart shows the HbA1c level among diabetes status, ranging from 3.5 to 9. It is noticeable that people without diabetes are primarily found in the lower HbA1c range (3.5 to 6.6), with counts continuously exceeding 7,000 per level. On the other hand, people with diabetes start to show up at HbA1c levels of 5.7 and above. Majority of people with diabetes has HbA1c level at 5.7, which sums up to 704 people. This suggests that higher HbA1c levels are strongly associated with the presence of diabetes, while lower levels are typical among non-diabetic individuals.

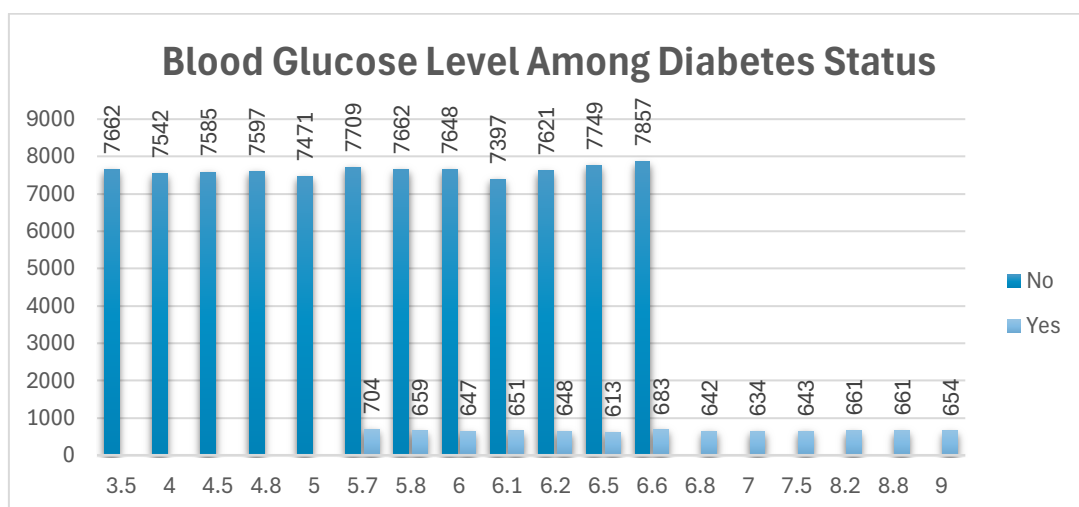


Figure 3.12 Blood Glucose Level Among Diabetes Status

The bar chart titled “Blood Glucose Level Among Diabetes Status” presents the distribution of patients across various blood glucose level. The lower blood glucose range (3.5 to 6.6) is dominated by individuals without diabetes, with consistently high counts exceeding 7,400 per level. On the other hand, people with diabetes start to show up at blood glucose levels 5.7 and higher, peaking between 613 to 704 people between levels 5.7 to 9. Overall, the chart shows a distinct shift in population distribution as blood glucose rises, supporting the strong correlation between high blood glucose levels and diabetes.

- 3) How does the prevalence of hypertension differ between diabetic and non-diabetic patients?

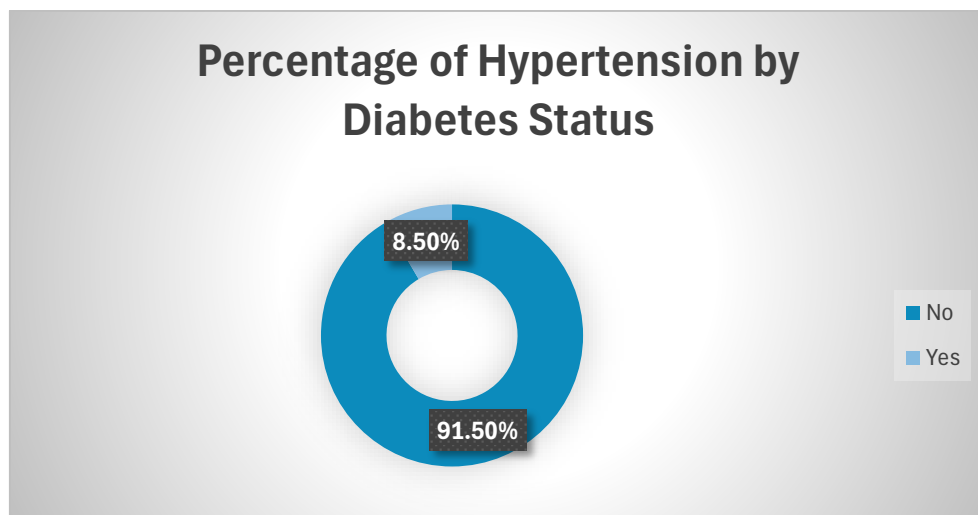


Figure 3.13 Percentage of Hypertension by Diabetes Status

The donut chart shows the proportion of hypertension patients among diabetic and non-diabetic groups. According to the chart, only 8.5% of hypertensive patients have diabetes while the other 91.5% do not. The majority of people with high blood pressure in this dataset do not have diabetes, despite the fact that hypertension and diabetes are frequently associated. The low percentage of diabetic hypertensive patients could be due to different risk profiles.

- 4) How does the prevalence of heart disease differ between diabetic and non-diabetic patients?

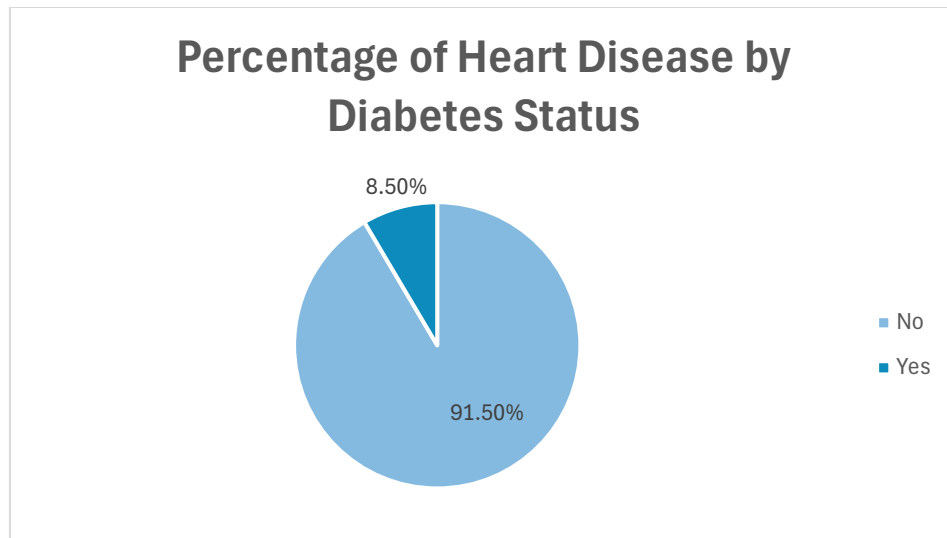


Figure 3.14 Percentage of Patients with Heart Disease by Diabetes Status

The pie chart shows the proportion of heart disease patients among diabetic and non-diabetic groups. It shows a significant difference in the rate between both groups. 8.5% of people without diabetes have heart disease, compared to 91.5% who do not. This implies that heart disease is more common in the diabetic group and a strong correlation between increased cardiovascular risk and diabetes.

5) How does smoking history differ between diabetic and non-diabetic patients?

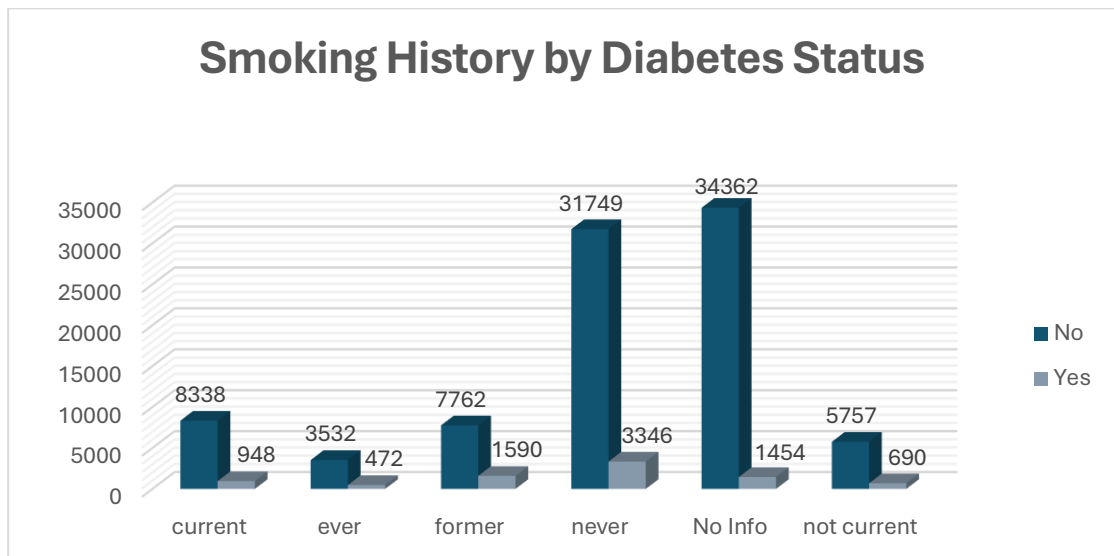


Figure 3.15 Status of Diabetes by Smoking History

The Smoking History by Diabetes Status chart compares the prevalence of diabetes among smoking history categories. There is a clear contrast between the two groups. The never and No Info categories comprise the majority of both diabetics and non-diabetics, with 31,749 and 34,362 non-diabetics and 3,346 and 1,454 diabetics respectively. Moreover, the former and current smoker categories have higher number of diabetics compared to their overall counts (1,590 and 948 respectively), which may indicate a connection between the prevalence of diabetes and smoking in the past or present. In contrast, the counts for the ever and not current categories tend to be lower. This distribution suggests that smoking history and diabetes risk may be related, especially for those who have smoked in the past or are currently smokers.

3.6.3 OBJECTIVE 3

1) How does diabetes prevalence vary across different age groups?

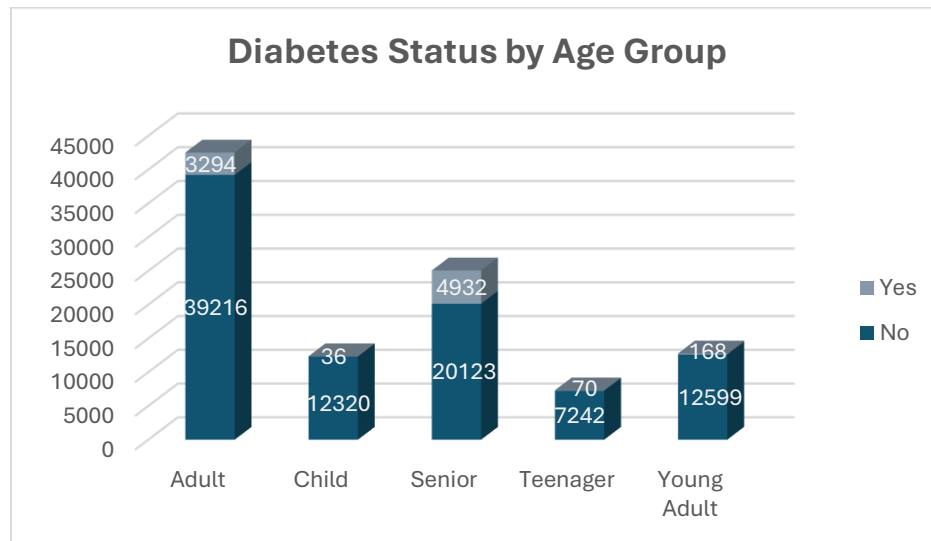


Figure 3.16 Diabetes Status by Age Group

The Diabetes Status by Age Group bar graph shows that the prevalence of diabetes varies significantly among age groups. With 3,294 and 4,932 cases respectively, adults and seniors have the highest numbers of people with diabetes, indicating that diabetes is more prevalent in older people. Children, teenagers and young adults, on the other hand, show significantly lower number of diabetes patients, just 36, 70 and 168 cases respectively. The number of people without diabetes is high across all age groups especially among adults (39,216) and children (12,320).

- 2) How has the distribution of diabetes cases across BMI categories changed from 2015 to 2022?

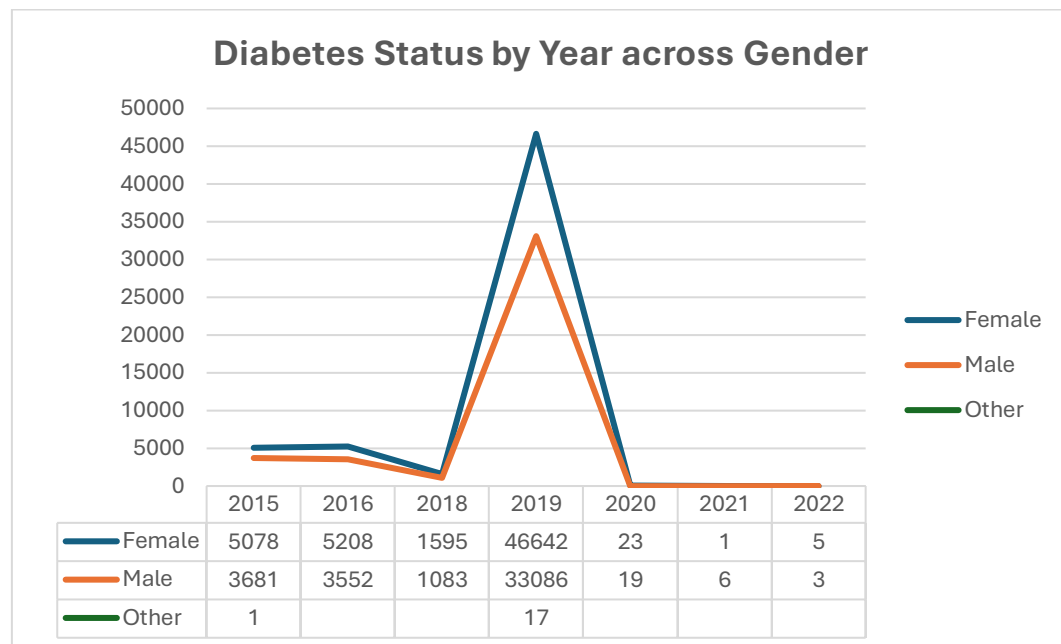


Figure 3.17 Diabetes Status by Year Across Gender

The line chart shows the diabetes status by year across gender from 2015 to 2022. The number of diabetes cases significantly across all gender categories starting 2018. Male and female cases peaked at 33,086 and 46,642 respectively, with 17 cases in the “Other” gender category, much more than in any previous year. On the other hand, majority of them has less than 6,000 cases per gender and almost none for “Other” from 2015 to 2018. After 2019, the numbers of diabetes across gender started to decline with the female having the highest number of diabetes (23) in 2020 while the “Other” category has no cases of diabetes after the year 2019.

3) How has the number of diabetes cases changed from 2015 to 2022 across gender categories?

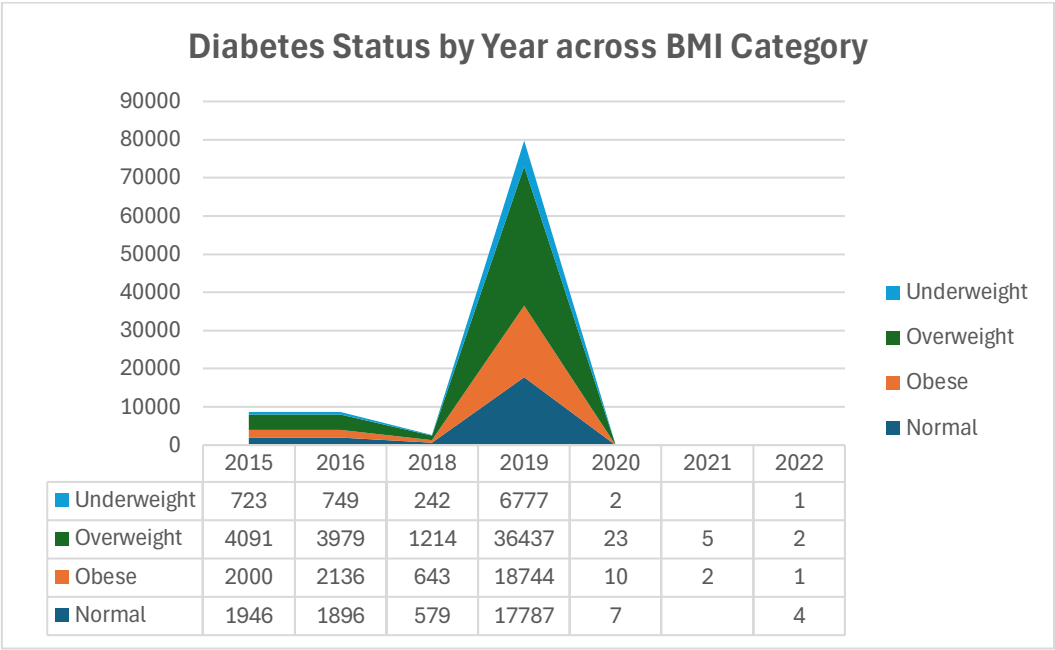


Figure 3.18 Diabetes Status by Year Across BMI Category

The line chart shows the diabetes status by year across BMI category. All BMI categories saw comparatively stable diabetes cases between 2015 and 2017, with overweight and obese people continuously exhibiting higher counts than those with normal and underweight BMIs. There was a remarkable increase in 2018, with 36,437 cases of overweight people and 18,744 cases of obese people, much higher than in previous years. Nonetheless, in 2019, all categories saw a sharp decline to single-digit counts, which continued until 2022.

3.7 DERIVATIVES ATTRIBUTES

There are 3 attributes had been derived for this project which are age group, BMI category and race. The Table 3.3 shows derivatives attributes for each attribute as shown as below:

Table 3.3 Derivatives Attributes

No	Variable Name	Variable Description	Data Types	Level of Measurement
1	Age Group	Grouped age range categories	Categorical	Ordinal
2	BMI Category	Classification of BMI	Categorical	Ordinal
3	Race	The race or ethnicity of the individual.	Categorical	Nominal

3.7.1 AGE GROUP

Categorical attribute that represents a patient's age range age rather than their exact age is derived by age group. It is derived to ease the user to identify the patient's age based on the age classification which are child, teenager, young adult, adult and senior. The function that is used to derive this attribute is:

=IF (D2<13, "Child", IF(D2<20, "Teenager", IF(D2<30, "Young Adult", IF(D2<60, "Adult", "Senior"))))

Where D2 refers to the first cell of 'Age'

	A	B	C	D
1	Year	Gender	Age Group	Age
2	2020	Female	Adult	32
3	2015	Female	Young Adult	29
4	2015	Male	Teenager	18
5	2015	Male	Adult	41
6	2016	Female	Adult	52
7	2016	Male	Senior	66
8	2015	Female	Adult	49
9	2016	Female	Teenager	15
10	2016	Male	Adult	51
11	2015	Male	Adult	42
12	2016	Male	Teenager	15
13	2016	Female	Adult	53
14	2015	Female	Child	3
15	2016	Female	Adult	40
16	2016	Female	Senior	64
17	2015	Female	Young Adult	23
18	2015	Female	Child	2
19	2015	Male	Senior	80
20	2016	Female	Child	12

Figure 3.19 Derivative of Age Group Attribute

3.7.2 BMI CATEGORY

A categorical attribute derived from the Body Mass Index (BMI) value is derived by BMI category. Based on the patient's body weight relative to height, it classifies them for analysing patients who may related to diabetes risk. The classifications are underweight, normal, overweight and obese. The function that is used to derive this attribute is:

=IF (K2<18.5,"Underweight", IF (K2<25,"Normal", IF(K2<30,"Overweight", "Obese"))))

Where K2 refers to the first cell of 'BMI'

J2				
=IF(K2<18.5,"Underweight", IF (K2<25,"Normal", IF(K2<30,"Overweight", "Obese")))				
	H	I	J	K
1	Heart Disease	Smoking History	BMI Category	BMI
2		never	Overweight	27.32
3		never	Normal	19.95
4		never	Normal	23.76
5		never	Overweight	27.32
6		never	Normal	23.75
7		not current	Overweight	27.32
8		current	Normal	24.34
9		No Info	Normal	20.98
10		never	Obese	38.14
11		No Info	Overweight	27.32
12		No Info	Normal	19.15
13		never	Obese	34.3
14		No Info	Normal	20.28
15		never	Overweight	27.63
16		ever	Obese	49.27
17		never	Overweight	27.32
18		never	Overweight	27.32
19		No Info	Overweight	27.32
20		never	Normal	20.78

Figure 3.20 Derivative of Age Group Attribute

3.7.3 RACE

A categorical demographic attribute that classifies ethnicity of the patients. This attribute is classified into a single race attribute allows users to immediately understand a patient's racial classification without interpreting multiple binary columns. These attributes are African American, Asian, Caucasian Hispanic and others.

F2				Other
		D	E	F
1	Age	Location		Race
2		32 Alabama		Other
3	ult	29 Alabama		Asian
4		18 Alabama		Other
5		41 Alabama		Caucasian
6		52 Alabama		African American
7		66 Alabama		Caucasian
8		49 Alabama		Caucasian
9		15 Alabama		Other
10		51 Alabama		African American
11		42 Alabama		Caucasian
12		15 Alabama		African American
13		53 Alabama		Caucasian
14		3 Alabama		African American
15		40 Alabama		Hispanic
16		64 Alabama		Other
17	ult	23 Alabama		Caucasian
18		2 Alabama		Asian
19		80 Alabama		Other
20		12 Alabama		Hispanic

Figure 3.21 Derivative of Race Attribute

3.8 DATA ISSUES

Although the dataset used in this study is complete and suitable for analysis, several limitations and challenges were identified during data interpretation and analysis. One of the main challenges encountered relates to tool performance constraints. The analysis involves a large dataset with multiple interactive visuals, calculated measures and filters developed using Excel and Power BI. As a result, performance issues such as slower data loading, refreshing or interaction may occur which can affect dashboard responsiveness during analysis.

Another limitation of this study is the lack of contextual variables within the dataset. While the dataset contains important demographic, lifestyle and clinical health indicators, it does not include additional factors such as diet, physical activity, income level or family medical history. These factors may also influence diabetes risk, and their absence limits the scope of interpretation by restricting the analysis to the available variables only.

In addition, categorisation through derived variables presents a further challenge in data interpretation. Variables such as age group and BMI category were derived to support clearer analysis and effective dashboard visualisation. Although this approach improves interpretability and comparison across groups, it involves grouping continuous data into categories which may reduce data granularity and limit detailed interpretation of individual level variants.

4.0 DASHBOARD PROTOTYPE

4.1 THE DASHBOARD

The Dashboard is developed to support the analysis of diabetes cases in United States. The dashboard consists of two main pages which are patient demographic and clinical and lifestyle comparison, each designed to address specific research objectives through visual exploration. Dashboard Page 1 focuses on patient demographic and distribution patterns, allowing users to examine diabetes cases across different age groups, genders, races, geographic locations and associated health conditions. Dashboard Page 2 emphasises clinical and lifestyle comparisons between diabetic and non-diabetic patients, highlighting differences in key health indicators, co-morbidities and smoking history. Interactive filters and selection buttons are incorporated to enable users to explore multiple views of the data. The screenshots presented in this section illustrate selected dashboard views under different filter and button selections to demonstrate the functionality and layout of the dashboard.

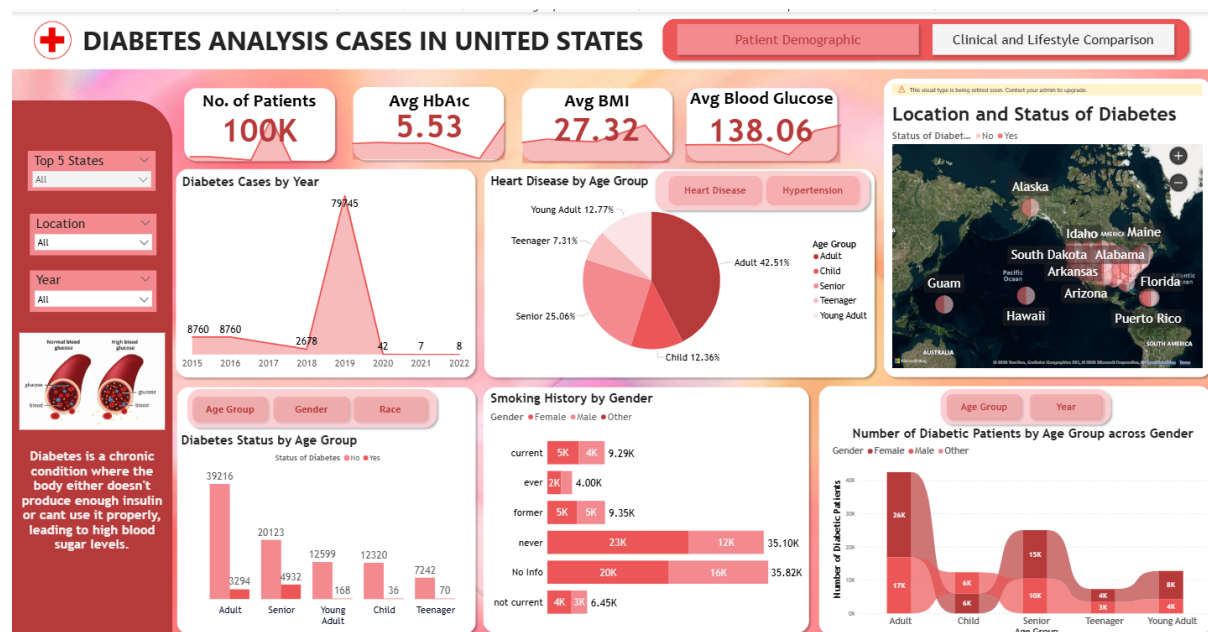


Figure 4.1 Dashboard Page 1 – View 1 (Age Group, Heart Disease buttons selected)

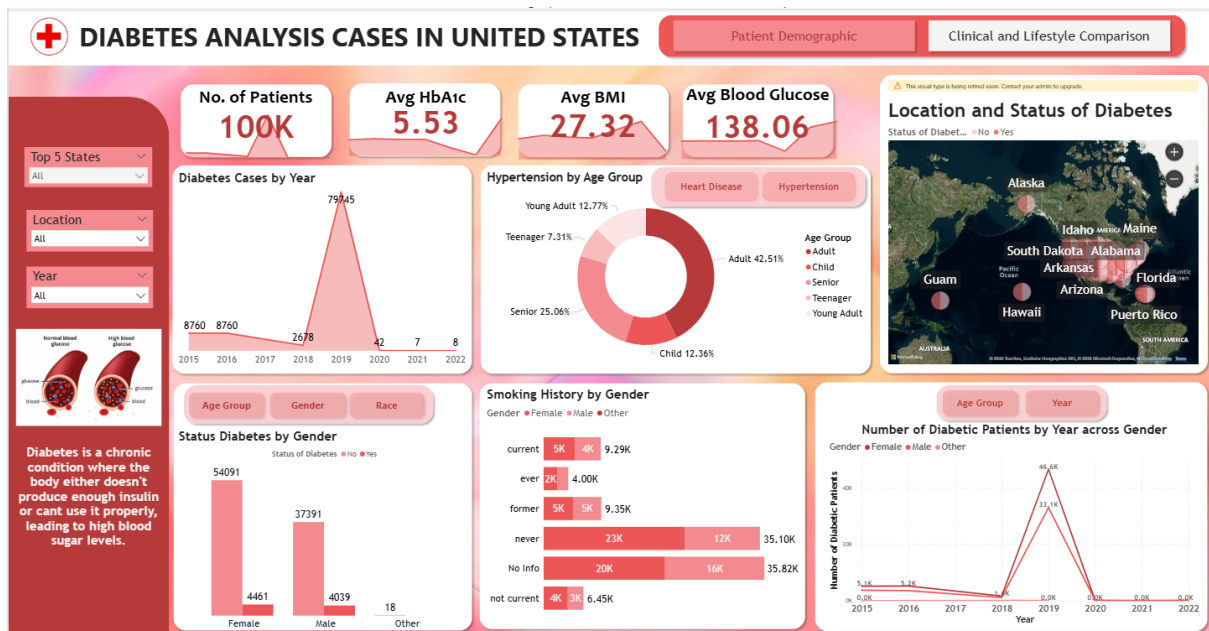


Figure 4.2 Dashboard Page 1 – View 2 (Gender, Hypertension and Year buttons selected)

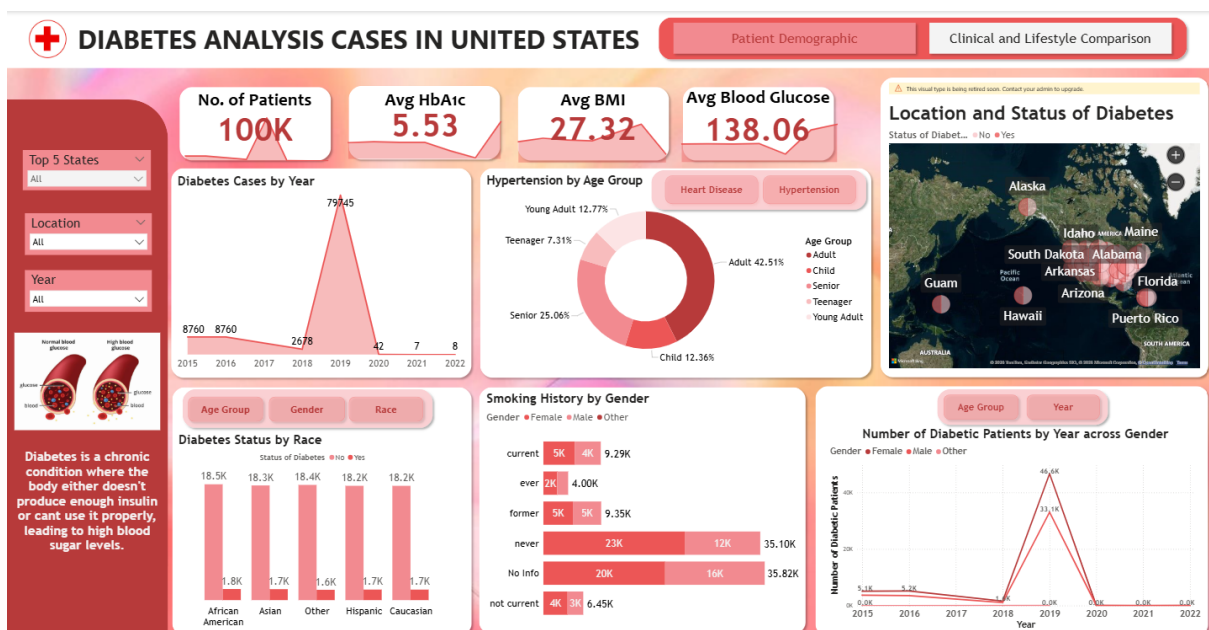


Figure 4.3 Dashboard Page 1 – View 3 (Race buttons selected)

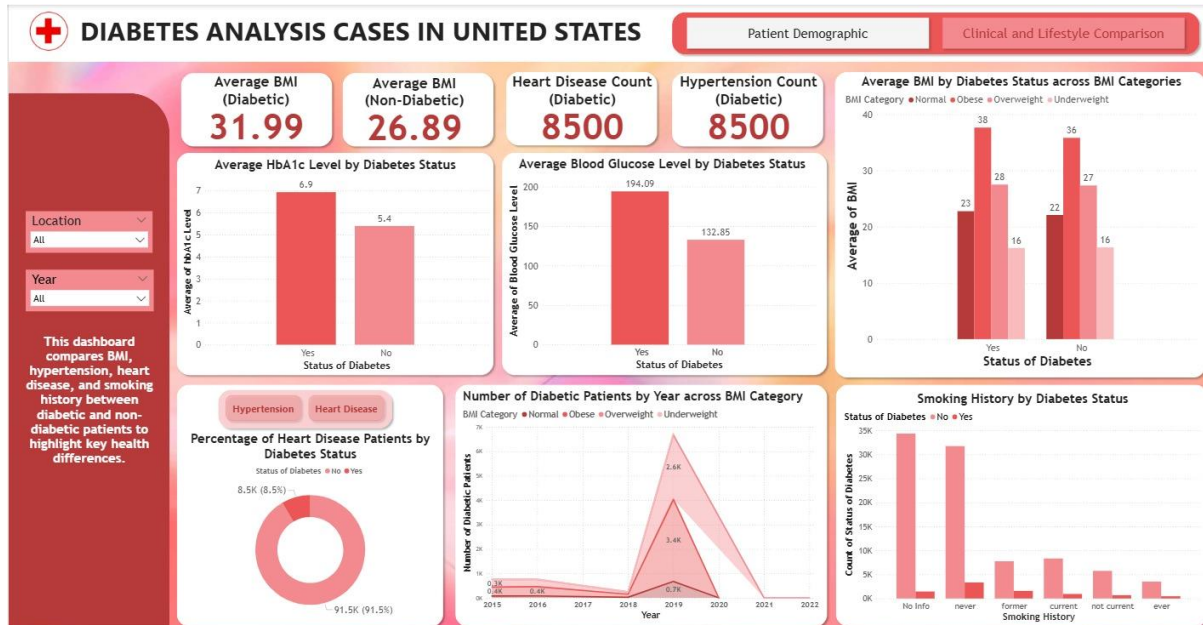


Figure 4.4 Dashboard Page 2 – View 1 (Heart Disease buttons selected)

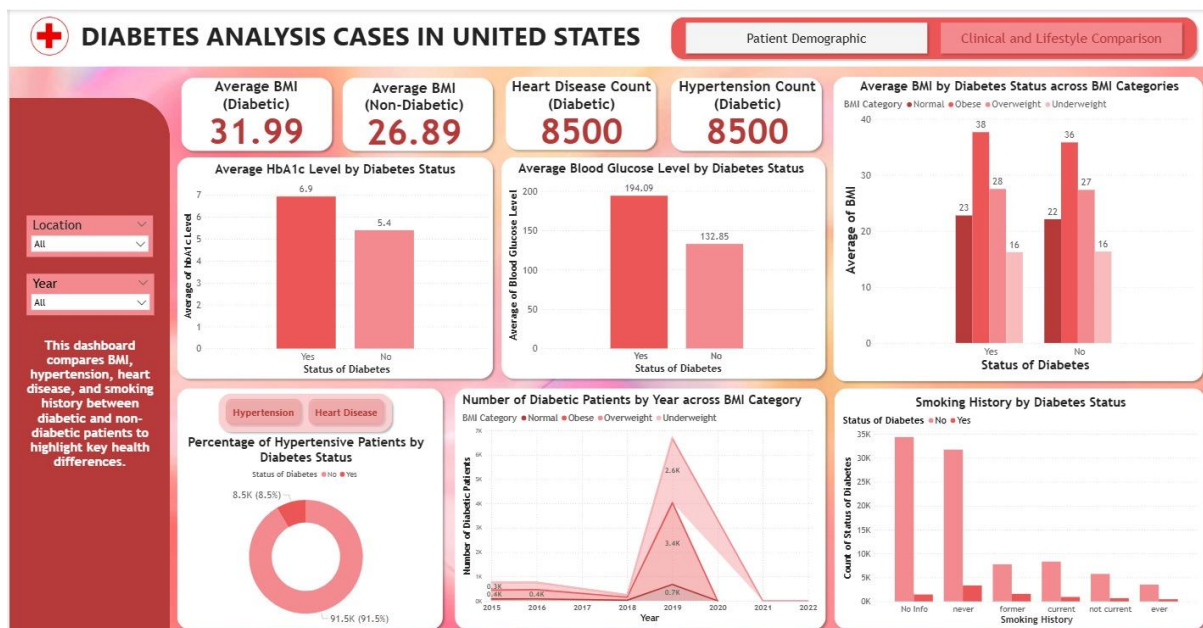


Figure 4.5 Dashboard Page 2 – View 2 (Hypertension buttons selected)

4.2 DASHBOARD ELEMENTS

This section presents the key elements incorporated within the interactive dashboard developed for this project. Each dashboard page consists of multiple visual and interactive components designed to support data exploration and address the defined research objectives. The elements are systematically numbered and labelled to clearly illustrate their function and placement within the dashboard layout. Dashboard Page 1 focuses on patient demographic and distribution analysis, while Dashboard Page 2 emphasises clinical and lifestyle comparisons between diabetic and non-diabetic patients. The annotated figures shown in this section highlight the individual dashboard elements under different views and filter selections, providing a clear reference for understanding how each component contributes to data navigation, visual interpretation and user interaction.

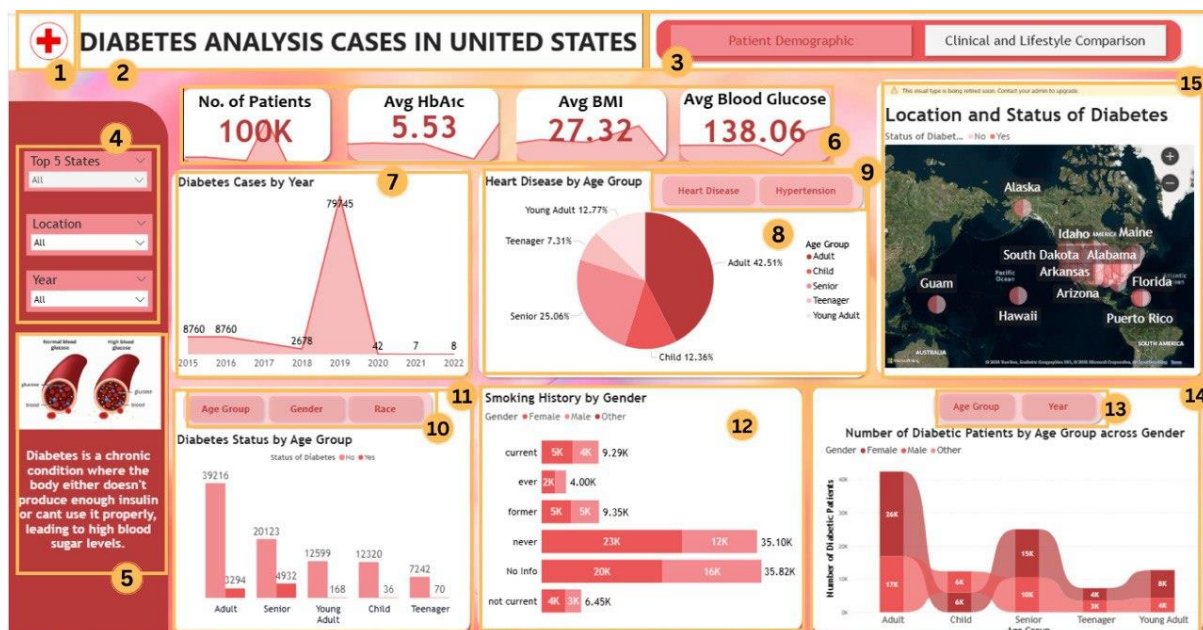


Figure 4.6 Elements of Dashboard Page 1 – View 1

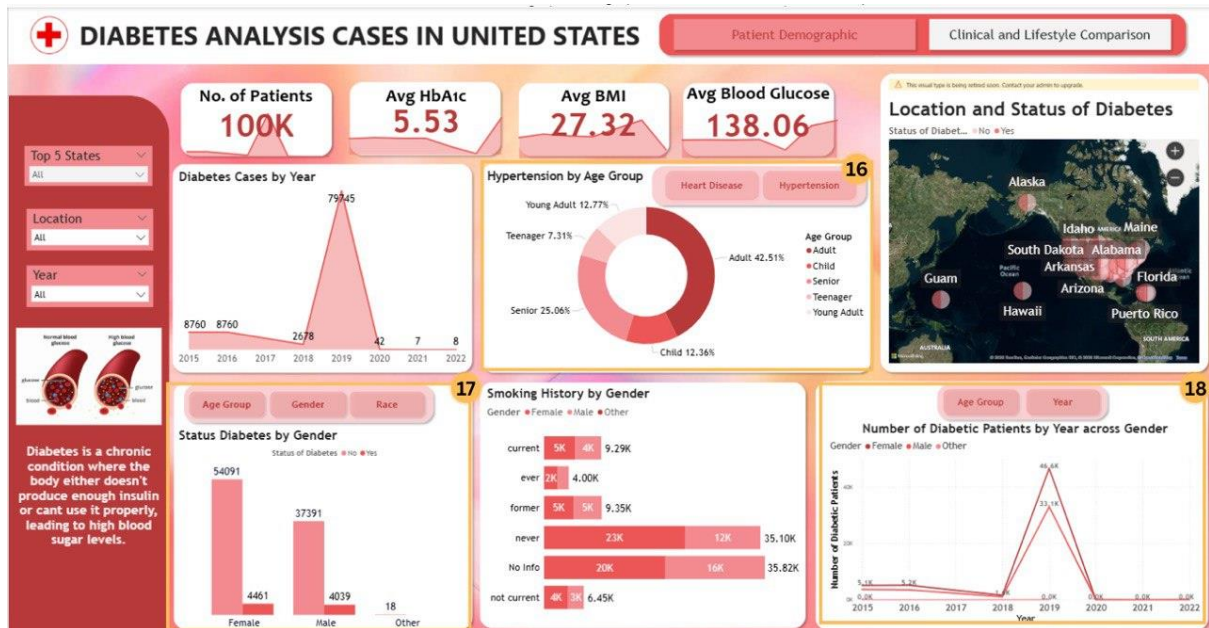


Figure 4.7 Elements of Dashboard Page 1 – View 2

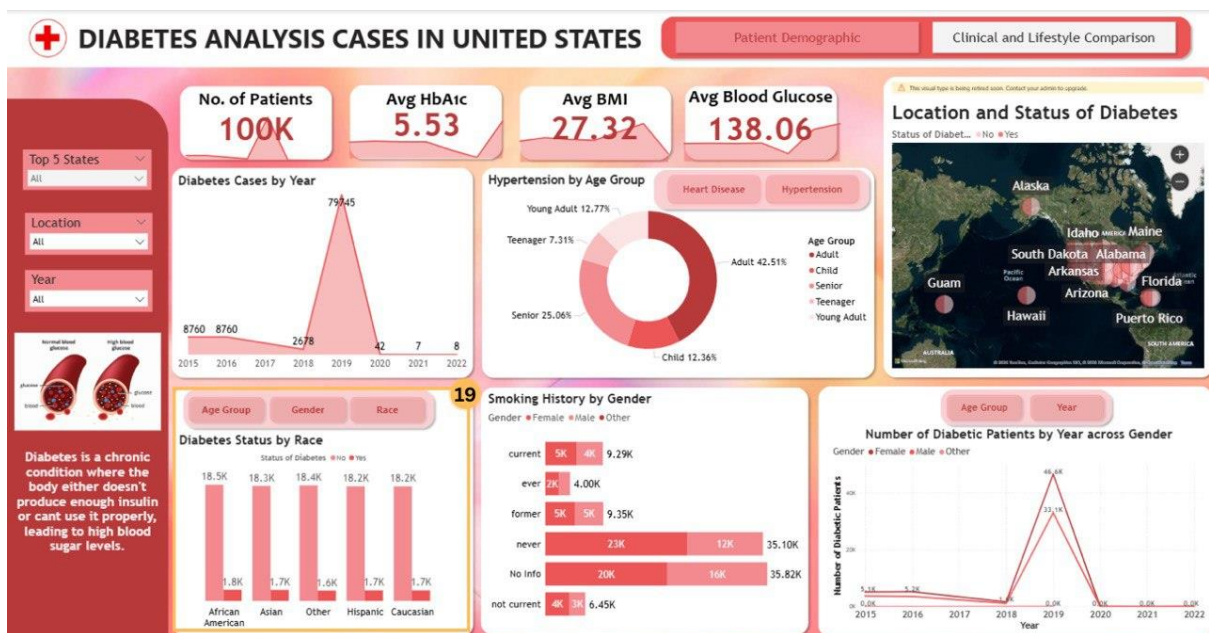


Figure 4.8 Elements of Dashboard Page 1 – View 3

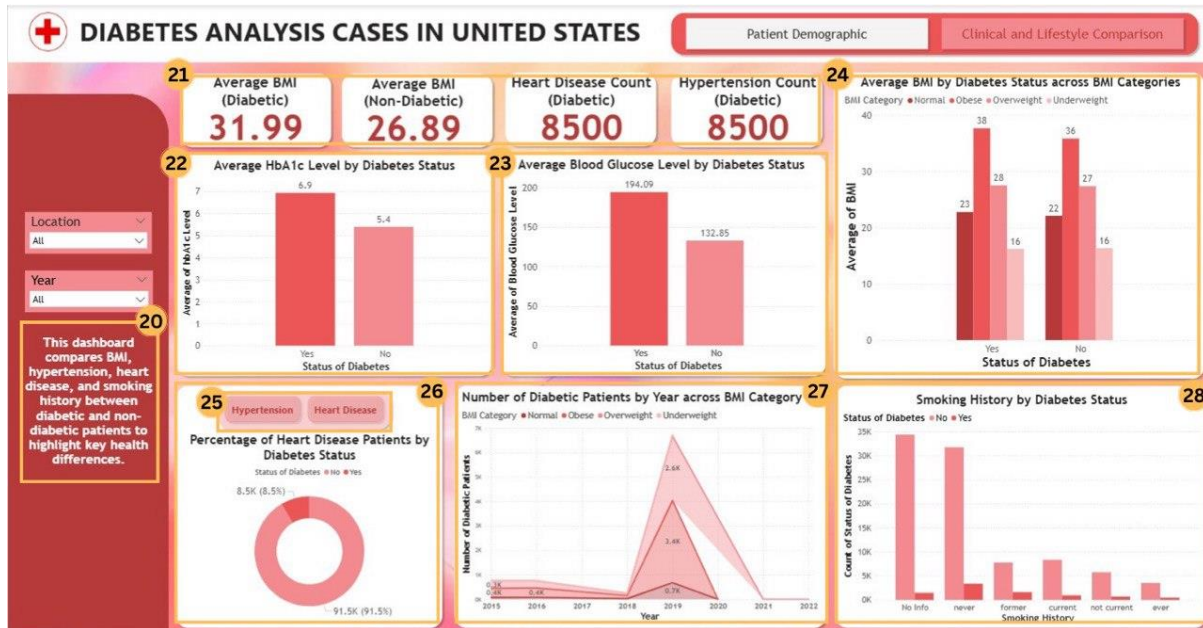


Figure 4.9 Elements of Dashboard Page 2 – View 1

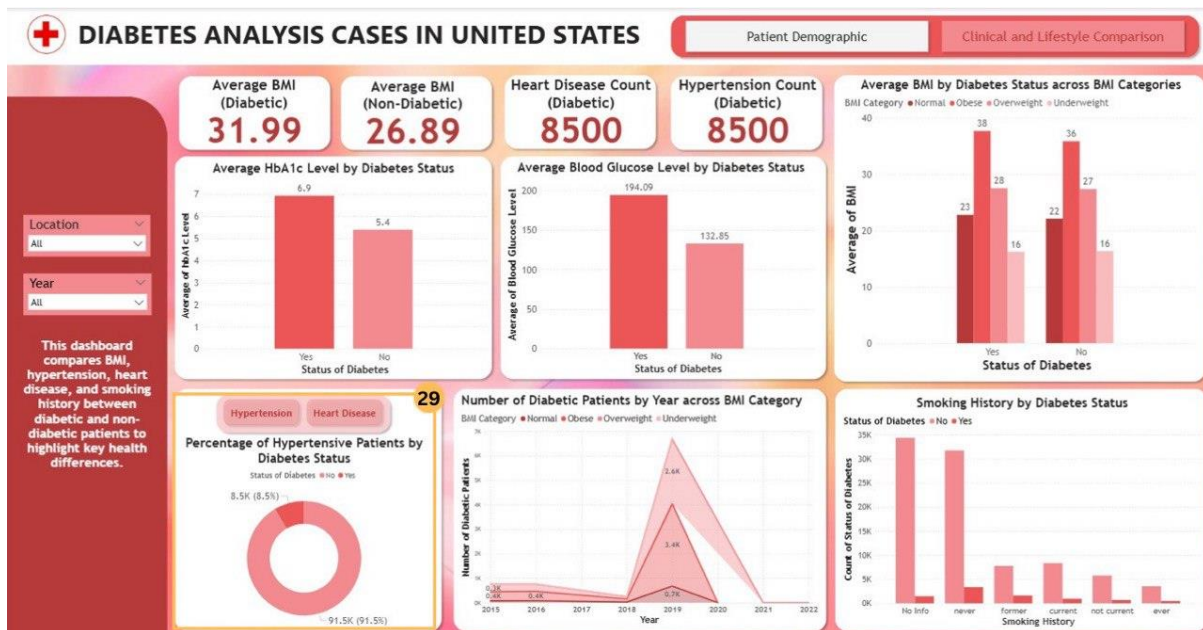


Figure 4.10 Elements of Dashboard Page 2 – View 2

Table 4.1 Description of the Elements of Dashboard Page 1 and Page 2

Label	Description
1	Dashboard icon
2	Title of the dashboard
3	Navigation tabs of the dashboard
4	Data filter panel
5	Diabetes information panel

6	Card for displaying the total patients, average HbA1c, average BMI, average Blood Glucose
7	Line chart of diabetes cases by year
8	Pie chart of heart disease by age group
9	Chart selection buttons: Heart Disease & Hypertension
10	Chart selection buttons: Age Group, Gender & Race
11	Bar chart of diabetes status by age group
12	Stacked bar chart of smoking history by gender
13	Chart selection buttons: Age Group & Year
14	Ribbon chart of diabetic patients by age group across gender
15	Map of diabetes status by location
16	Donut chart of hypertension by age group
17	Bar chart of diabetes status by gender
18	Line chart of diabetes patients by year across gender
19	Bar chart of diabetes status by race
20	Dashboard description panel
21	Card for displaying the average BMI for diabetic patients, average BMI for non-diabetic patients, heart disease counts, hypertension counts
22	Bar chart of average HbA1c Level by diabetes status
23	Bar chart of average blood glucose level by diabetes status
24	Bar chart of average BMI by diabetes status across BMI category
25	Chart selection buttons: Hypertension & Heart Disease
26	Donut chart of heart disease percentage by diabetes status
27	Line chart of diabetic patients by year across BMI category
28	Bar chart of smoking history by diabetes status
29	Donut chart of hypertension percentage by diabetes status

4.3 DASHBOARD ARRANGEMENTS

This section illustrates the arrangement and reading flow of the dashboard components based on the defined research objectives. The dashboard is designed to guide users through a logical sequence of visual analysis, ensuring that insights are explored in an organised and objective-driven manner. Charts and visual elements are grouped and arranged according to

research objectives to support intuitive navigation and interpretation. The figures presented in this section demonstrate how users are guided from one group of visuals to another, highlighting the transition between demographic analysis, clinical and lifestyle comparisons and temporal trend exploration across the dashboard pages.

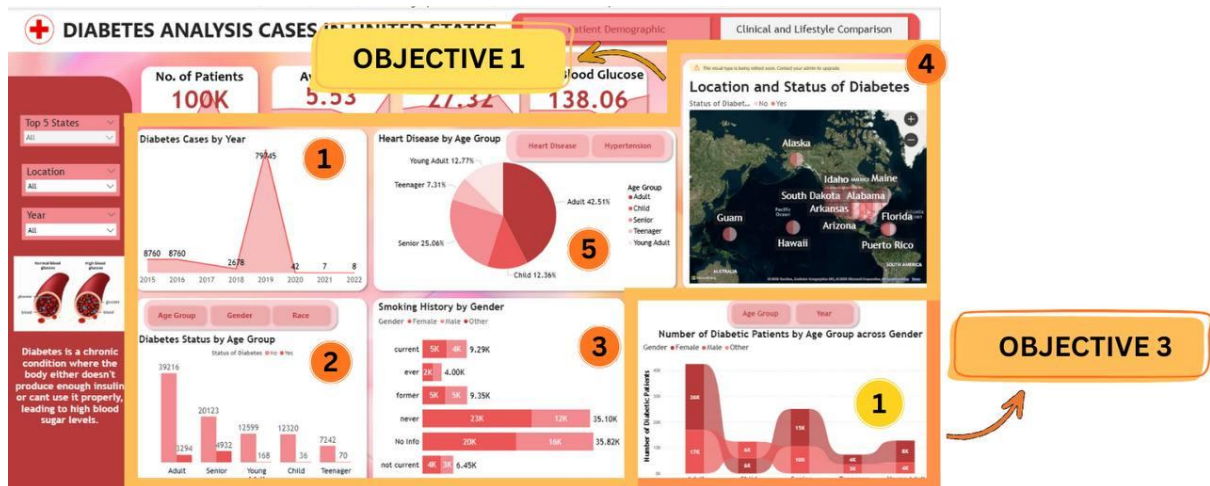


Figure 4.11 Reading Flow and Objectives of Dashboard Page 1 – View 1

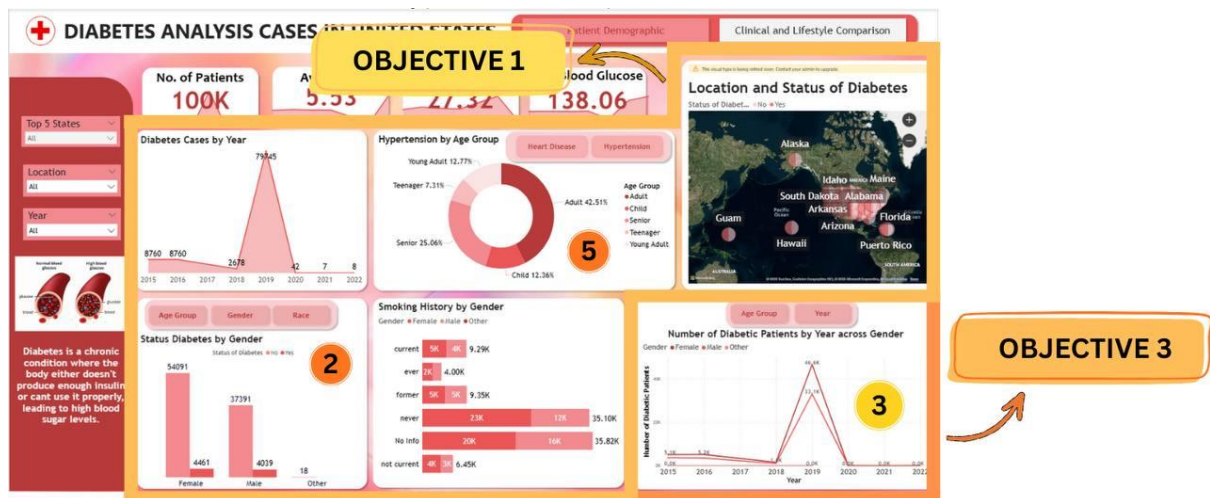


Figure 4.12 Reading Flow and Objectives of Dashboard Page 1 – View 2



Figure 4.13 Reading Flow and Objectives of Dashboard Page 1 – View 3

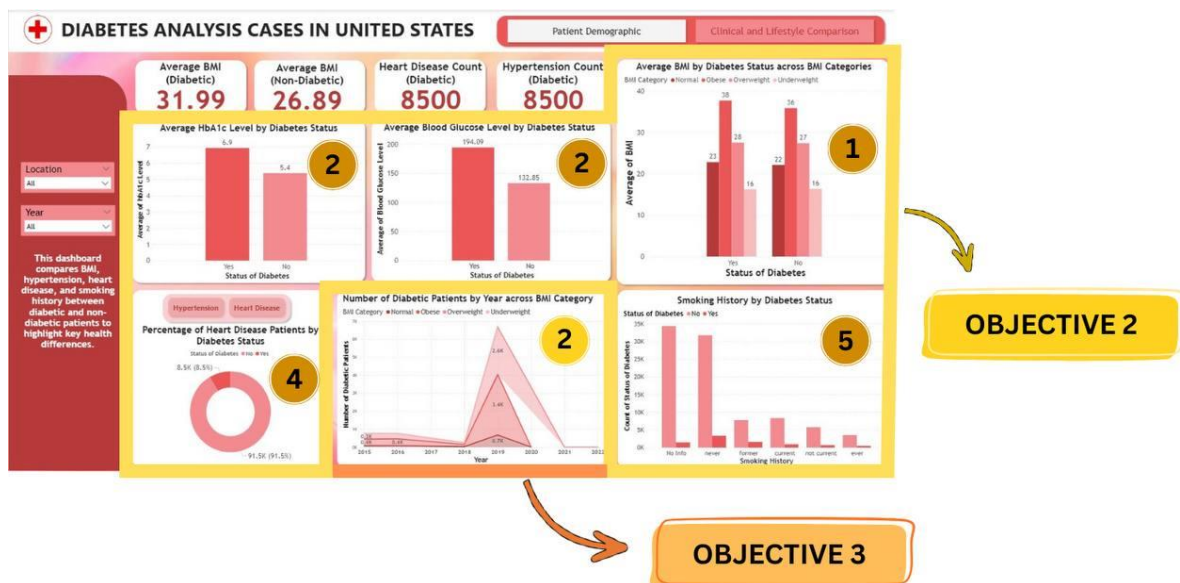


Figure 4.14 Reading Flow and Objectives of Dashboard Page 2 – View 1

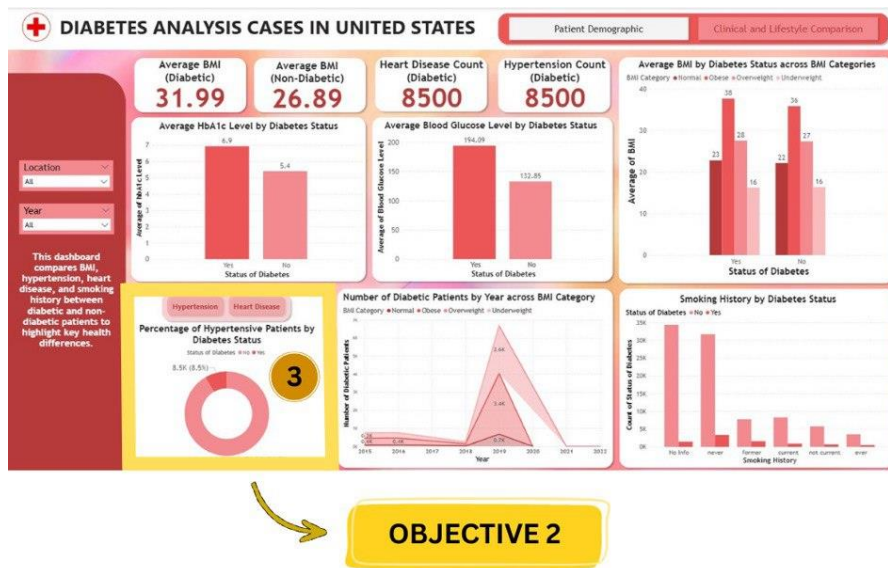


Figure 4.15 Reading Flow and Objectives of Dashboard Page 2 – View 2

4.4 DASHBOARD LINK

The interactive dashboard developed for this project can be accessed via the following link:

https://app.powerbi.com/links/YyMhXVOtX5?ctid=cdebb0e2-9fea-4f54-8670-672707797ada&pbj_source=linkShare&bookmarkGuid=4cbfd7f2-a6d5-4129-903f-20edfaa79b28

5.0 DATA INSIGHTS AND DISCUSSION

5.1 VISUAL DESCRIPTION AND PURPOSES

The Table 5.1 shows descriptions and purposes for each chart as shown as below:

Table 5.1 Charts Descriptions and Purposes

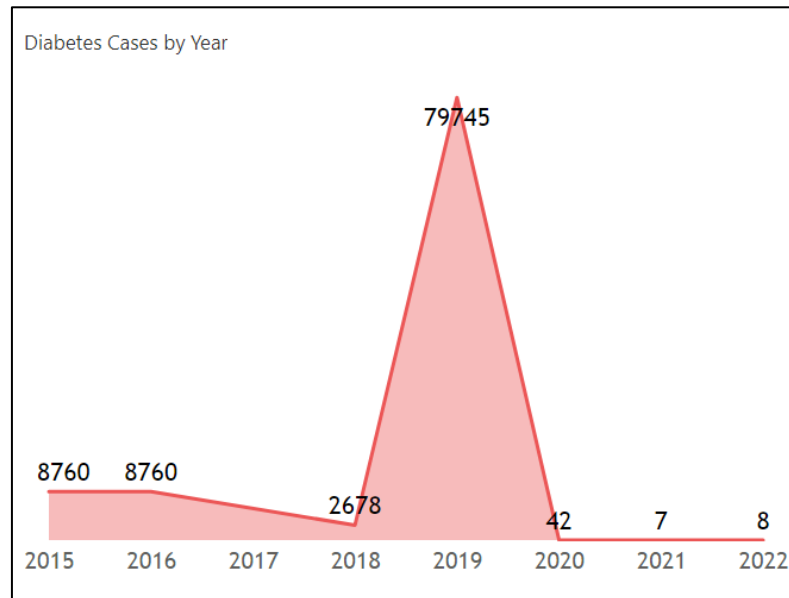
No	Title of Charts	Description
1	Diabetes Cases by Year	Uses two attributes which are year and number status of diabetes to display the distribution of diabetes cases over time from 2015 to 2022.
2	Diabetes Status by Age Group, Gender, and Race	Uses three demographic attributes which are age group, gender and race, together with diabetes status to display the distribution of diabetic and non-diabetic. Interactive buttons allow users to switch between age group, gender and race view for comparison of diabetes status.
3	Smoking History by Gender	Uses two attributes which are smoking history category and gender to display the distribution of patients across different smoking status by gender.
4	Location and Status of Diabetes	Uses two attributes which are geographic location and diabetes status to display the spatial distribution of diabetic and non-diabetic patients across different locations.
5	Heart Disease and Hypertension by Age Group	Uses three attributes which are heart disease, hypertension and age group to display the distribution of heart disease and hypertension across different age categories.
6	Average BMI by Diabetes Status across BMI Categories	Uses three attributes which are diabetes status, BMI value and BMI categories to display the comparison of average BMI across different BMI categories for diabetic and non-diabetic patients.
7	Average Blood Glucose and Average HbA1c Level by Diabetes Status	Uses two clinical measurement attributes which are average blood glucose level and average HbA1c level with diabetes status to display the comparison of

		glycaemic indicators between diabetic and non-diabetic patients.
8	Percentage of Hypertensive Patients by Diabetes Status	Uses two attributes which are diabetes status and percentage of hypertensive patients to display the proportion of hypertensive individuals within diabetic and non-diabetic groups.
9	Percentage of Heart Disease Patients by Diabetes Status	Uses two attributes which are diabetes status and percentage of heart disease patients to display the proportion of hypertensive individuals within diabetic and non-diabetic groups.
10	Smoking History by Diabetes Status	Uses two attributes which are smoking history category and diabetes status to display the distribution of diabetic and non-diabetic patients across different smoking history groups.
11	Number of Diabetes Patients by Year across BMI Category	Uses three attributes which are year, BMI category and number of diabetic patients to display the yearly distribution of diabetic patients across different BMI classifications.
12	Number of Diabetic Patients by Age Group and Year across Gender	Uses four attributes which are gender, age group, year and number of diabetic patients to display the distribution of diabetes cases across different genders.

5.2 STORYTELLING NARRATIVES

Research Objective 1: To identify demographic and health distribution patterns among patients.

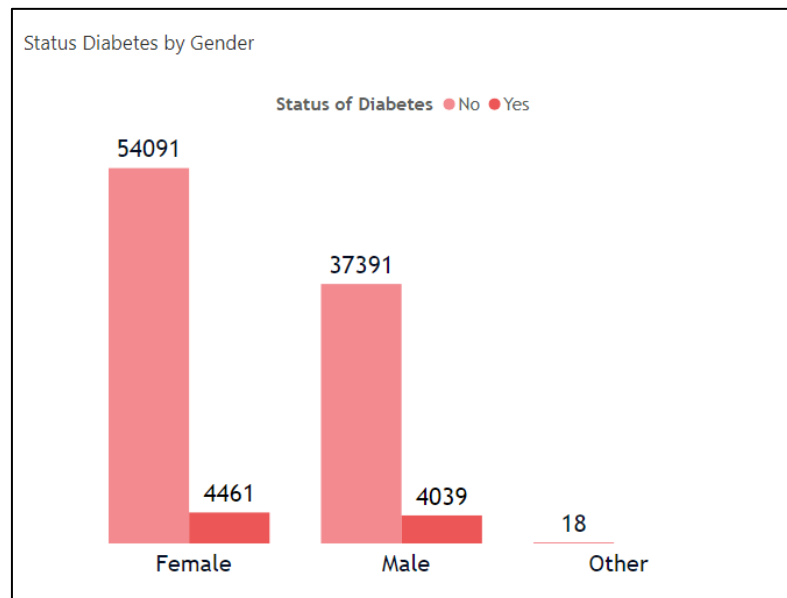
Research Question 1.1: What is the distribution and trend of diabetes status from 2015 to 2022?



RO1: Figure 1a Diabetes Cases by Year

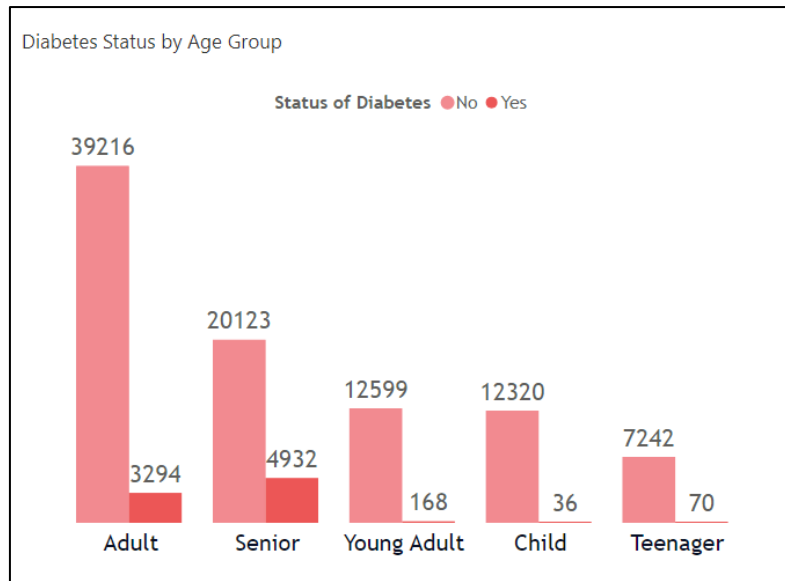
This line chart presents the annual trend of diabetes cases among patients from 2015 and 2022. The number of diabetes cases remained constant at 8,760 cases in both 2015 and 2016, followed by a decline to 2,678 cases in 2018. A sharp increase is observed in 2019, where diabetes cases peaked at 79,745 representing the highest value across the entire period. After this peak, the number of cases decreased substantially to 42 cases in 2020 and remained at very low levels in subsequent years with only 7 cases recorded in 2021 and 8 cases in 2022. Overall, the chart highlights 2019 as an outlier year with an exceptionally high number of diabetes cases compared to other years suggesting potential differences in data reporting or collection for that period.

Research Question 1.2: How are diabetic and non-diabetic patients distributed by gender, age group, and race?



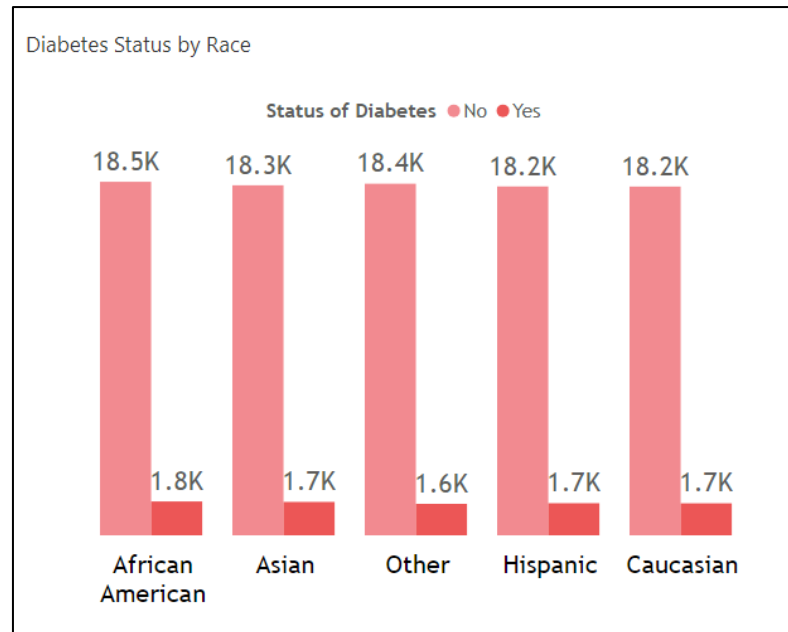
RO1: Figure 1b Status Diabetes by Gender

This clustered column chart displays the distribution of diabetes status across gender categories which are female, male and other. Overall, the chart shows that non-diabetic patients outnumber diabetic patients across both female and male groups, Females represent the largest group in the dataset with 54,091 non-diabetic patients and 4,461 diabetic patients. In comparison, males represent 37,391 non-diabetic patients and 4,039 diabetic patients. Although females have a slightly higher number of diabetes cases than males, the difference between the two groups is relatively small. The Other gender category shows a minimal presence in the dataset with only 18 non-diabetic records and no recorded diabetic cases. In summary, the chart indicates that females are more highly represented in the dataset for both diabetic and non-diabetic patients while the majority of patients across all gender groups do not have diabetes.



RO1: Figure 1c Diabetes Status by Age Group

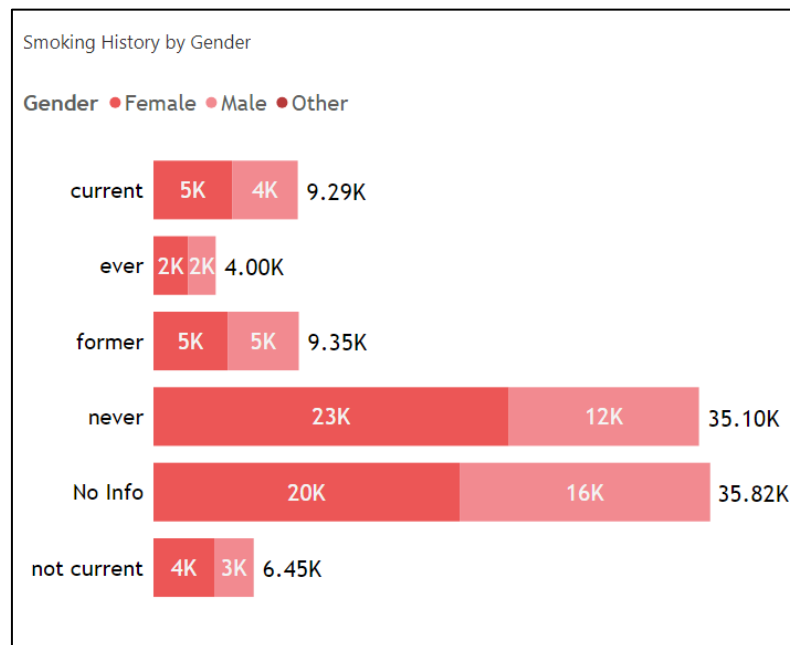
This clustered column chart presents the distribution of diabetes status across different age groups which are adults, seniors, young adults, children and teenagers. Adults represent the largest age group in the dataset with 39,216 non-diabetic individuals and 3,294 diabetic patients reflecting their dominant presence in the overall population. Although seniors form a smaller group overall, they have the highest number of diabetic patients at 4,932 compared to 20,123 non-diabetic individuals. In contrast, younger age groups show substantially lower numbers of diabetes cases with 168 cases among young adults, 70 cases among teenagers and 36 cases among children, despite having outstanding non-diabetic populations. Overall, the chart indicates that non-diabetic patients outnumber diabetic patients across all age groups and diabetes cases are more focused within the adult and senior populations in this dataset.



RO1: Figure 1d Diabetes Status by Race

This clustered column chart illustrates the distribution of diabetes status across different race categories including African American, Asian, Hispanic, Caucasian and Other. Overall, the chart shows that non-diabetic patients outnumber diabetic patients across all racial groups. Among diabetic patients, African Americans record the highest count approximately 1.8K cases followed closely by Asian, Hispanic and Caucasian groups, each with around 1.7K cases. The Other race category records a slightly lower number of diabetic cases at approximately 1.6K. Similarly, the number of non-diabetic patients is relatively consistent across all race groups, ranging between approximately 18.2K and 18.5K. In summary, the distribution of diabetes cases across race categories appears relatively similar in this dataset with no substantial differences observed in the counts of diabetic and non-diabetic patients among the different racial groups.

Research Question 1.3: Does smoking history differ by gender?



RO1: Figure 1e Smoking History by Gender

This stacked bar chart illustrates the distribution of smoking history categories by gender including current, ever, former, never, not current and records with no information. Overall, the never category represents the largest reported smoking status for both genders with a total of 35.10K individuals, consist of 23K females and 12K males who have never smoked. The no info category is also substantial represent 35.82K records which indicates a large number of patients with missing smoking history information in the dataset including 20K females and 16K males.

Among individuals with a smoking history, current smokers total 9.29K consisting of 5K females and 4K while former smokers represent for 9.35 cases and are evenly distributed between females and males at 5K each. The ever-smoking category records a smaller total of 4K cases with equal representation between females and males at 2K each. Additionally, the not current category includes 6.45K individuals with 4K females and 3K male. Overall, the chart indicates that non-smoking status never is the most common reported category while smoking related categories show relatively balanced distributions across genders within this dataset.

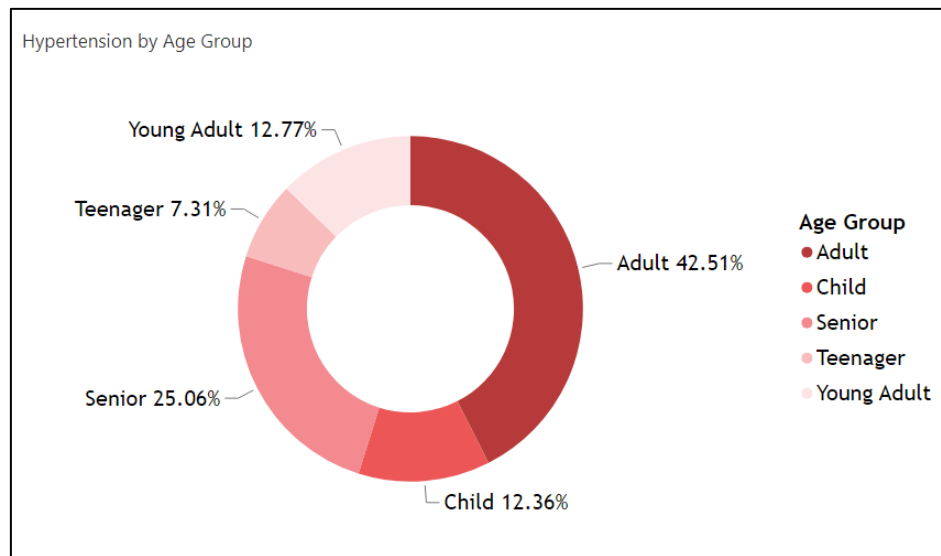
Research Question 1.4: How are diabetes cases geographically distributed across US states?



RO1: Figure 1f Location and Status of Diabetes

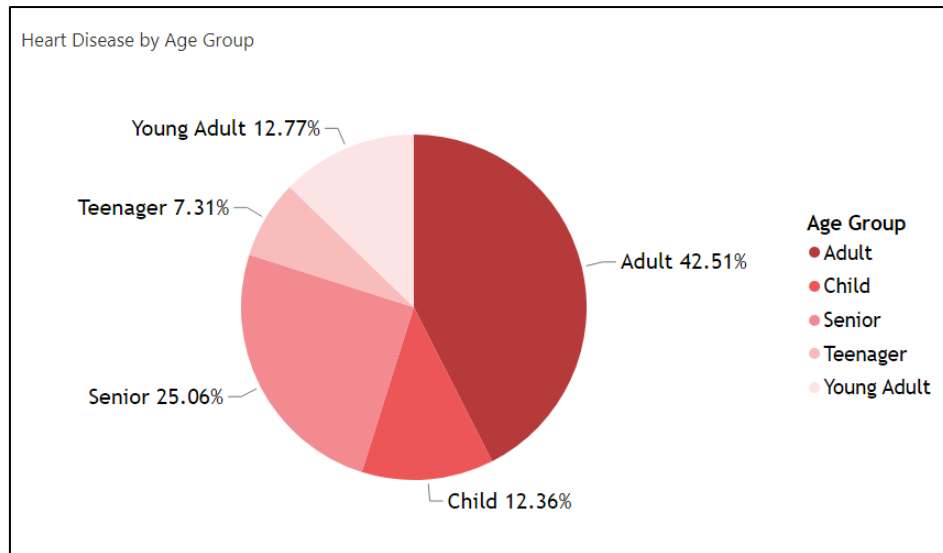
This map visualization illustrates the geographic distribution of diabetic and non-diabetic patient records across different locations in the United States and its states. Each location symbol represents the number of patients recorded in the dataset, with colour differentiation indicating diabetes status. The map allows for a longitudinal overview of how patient records are distributed across states such as Alaska, Alabama, Florida, Arizona, Hawaii, Guam and Puerto Rico. Overall, the visualization provides a clear geographic context for understanding where diabetes related records are concentrated within the dataset, without adjusting for population size.

Research Question 1.5: What is the proportion of patients with hypertension or heart disease among age groups?



RO1: Figure 1g Hypertension by Age Group

This donut chart illustrates the percentage distribution of hypertension cases across different age groups. Adults represent the largest proportion of hypertension cases at 42.51% followed by seniors at 25.06%. In comparison, younger age groups contribute smaller percentages, with young adults represent for 12.77%, children for 12.36% and teenagers for 7.31%. Overall, the chart indicates that hypertension cases are more focussed among adults and seniors, while younger age groups represent a smaller share of the total hypertension cases.

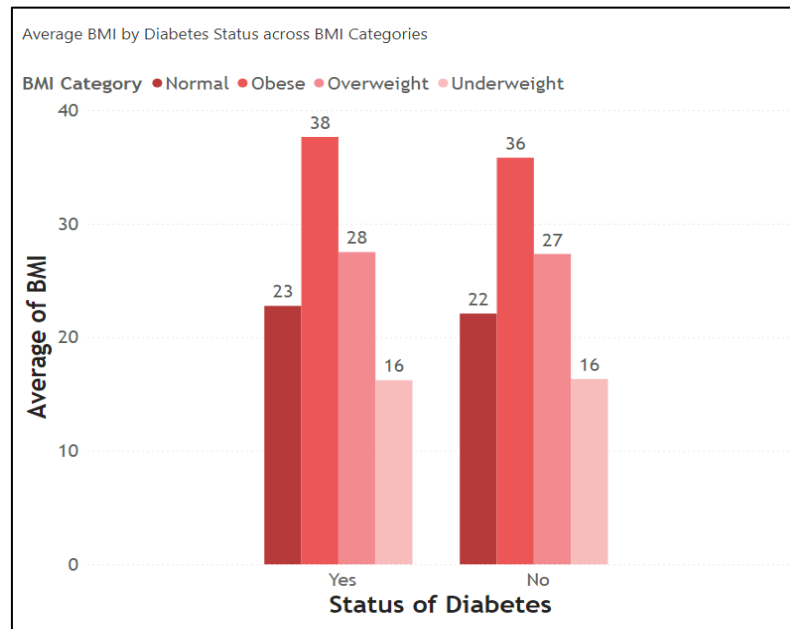


RO1: Figure 1h Heart Disease by Age Group

This pie chart illustrates the percentage distribution of diagnosed heart disease cases across different age groups. Adults represent for the largest proportion of heart disease cases at 42.51%, followed by seniors at 25.06%. Younger age groups contribute smaller shares of cases, with young adults representing 12.77%, children for 12.36% and teenagers 7.31%. Overall, the chart indicates that heart disease cases are more focused among adult and senior age groups, while younger populations represent a smaller proportion of the total cases.

Research Objective 2: To compare health risk indicators between diabetic and non-diabetic patients.

Research Question 2.1: How does average BMI differ between diabetic and non-diabetic patients across BMI categories?



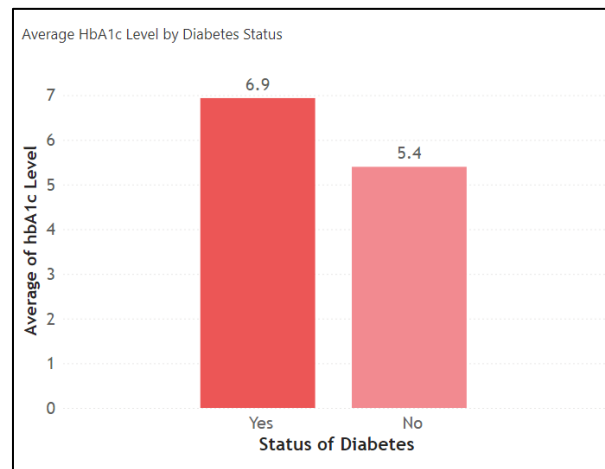
RO2: Figure 2a Average BMI by Diabetes Across BMI Categories

The analysis starts with the body mass index (BMI) in Figure 2a, which demonstrates that patients with diabetes frequently have higher average BMI values in practically every BMI category. Patients with diabetes have an average BMI of 38 in the obese category, while non-diabetic patients have a BMI of 36.

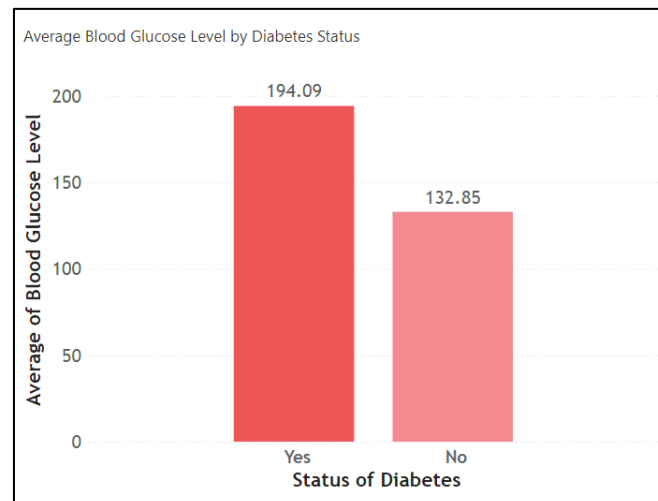
In the overweight category, diabetic patients have an average of 28, which is slightly higher than the average of 27 for non-diabetics. Diabetics patients have an average BMI of 23, which is higher than that of non-diabetic patients, even within the normal BMI range. The only exception is the underweight category, where the average BMI of both groups is 16.

These results demonstrate the strong correlation between diabetes risk and excess body weight, especially in the overweight and obese categories, supporting obesity as a major underlying factor rather than a coincidental condition.

Research Question 2.2: How does HbA1c level and blood glucose level differ between diabetic and non-diabetic patients?



RO2: Figure 2b Average HbA1c Level by Diabetes Status

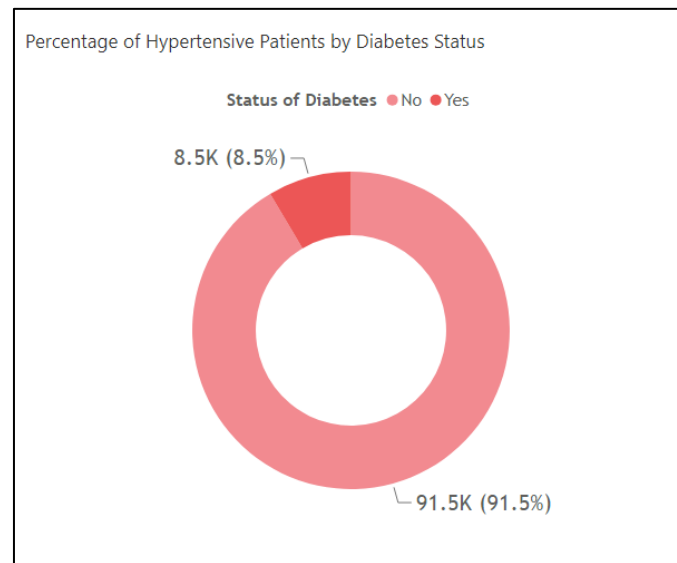


RO2: Figure 2c Average Blood Glucose Level by Diabetes Status

The biological differences between patients with and without diabetes are further demonstrated in Figure 2b and 2c. Patients without diabetes maintain an average HbA1c level of 5.4 in Figure 2b, while those with diabetes record a significantly higher average of 6.9. this 1.5-point difference validates HbA1c as a crucial marker for the diagnosis of diabetes and long-term glucose regulation.

In the similar hand, Figure 2c illustrates a significant disparity in average blood glucose levels. Patients without diabetes have an average blood glucose level of 132.85, whereas those with diabetes have an average blood glucose of 194.09, which is more than the non-diabetic value. These findings clearly show that diabetes is characterized by persistently high blood glucose levels.

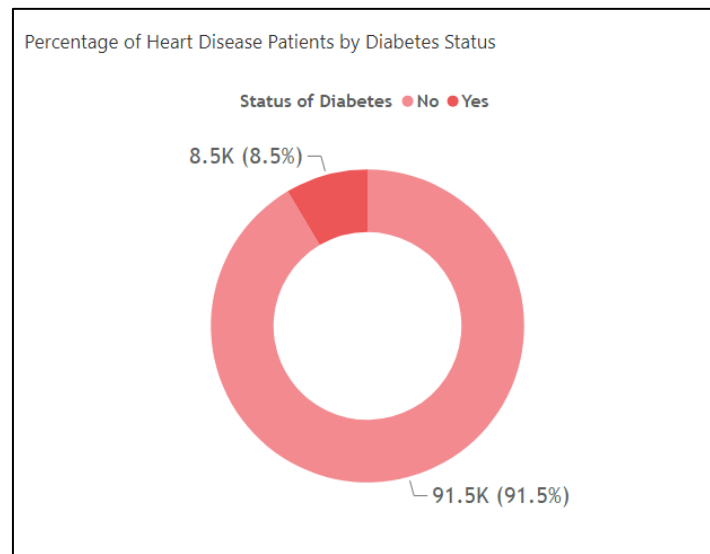
Research Question 2.3: How does the prevalence of hypertension differ between diabetic and non-diabetic patients?



RO2: Figure 2d Hypertension Patients by Diabetes Status

Figure 2d investigates the relationship between diabetes and hypertension. According to the figure, 8.5K patients (8.5%) in the dataset have both diabetes and hypertension. On the other hand, 91.5K patients (91.5%) do not have diabetes. The presence of hypertension among diabetic patients indicates a significant overlap between these two conditions, even though the non-diabetic group makes up the majority of the dataset. This implies that people with diabetes are more likely to have hypertension, highlighting the significance of routine blood pressure checks for diabetics.

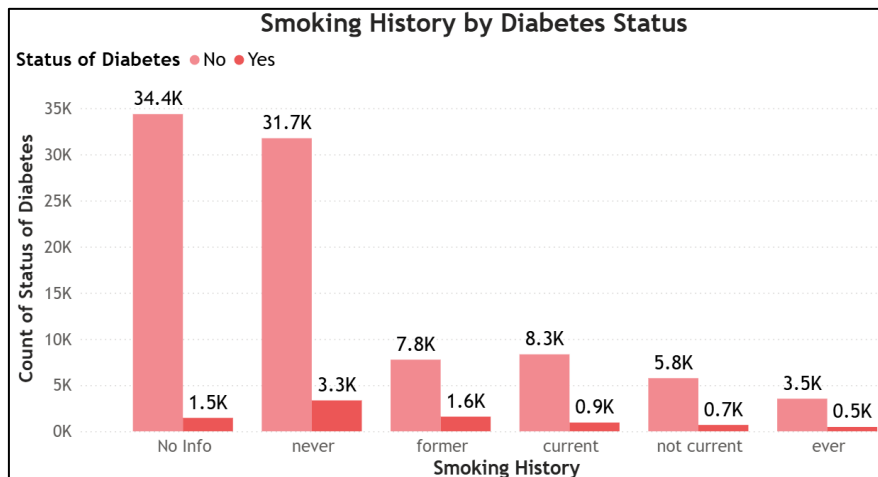
Research Question 2.4: How does the prevalence of heart disease differ between diabetic and non-diabetic patients?



RO2: Figure 2e Heart Disease by Diabetes Status

The figure above shows a similar pattern for heart disease. According to the chart, 91.5K patients (91.5%) do not have diabetes, while 8.5K patients (8.5%) have both diabetes and heart disease. This overlap shows that diabetes and cardiovascular problems are closely related. The results highlight the necessity of early cardiovascular monitoring among diabetic patients.

Research Question 2.5: How does smoking history differ between diabetic and non-diabetic patients?

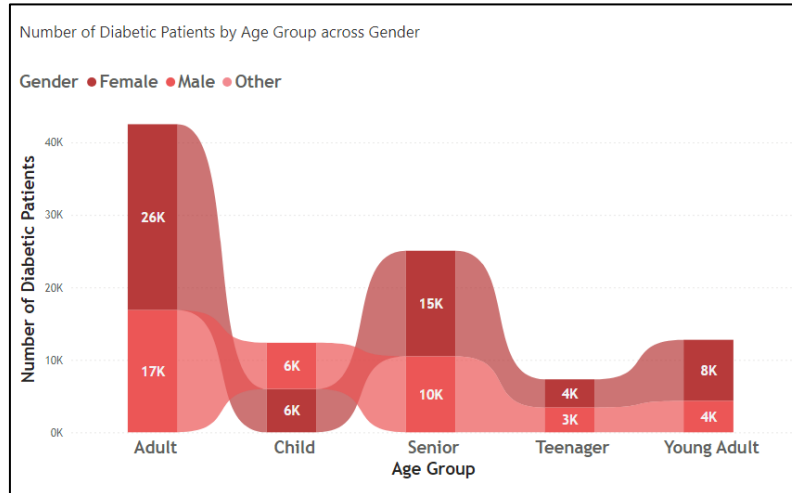


RO2: Figure 2f Smoking History by Diabetes Status

Lastly, smoking history across diabetes status is examined in Figure 2f. compared to 3.3K diabetic patients in the same category, about 31.7K non-diabetic patients say they have never smoked. However, there is a significant history of smoking among the diabetic group, with 1.6K former smokers and almost 1.0K current smokers. For both groups, a significant portion of records all into the “No Info” category, 34.4K patients without diabetes and 1.5K patients with diabetes. Despite the limitation, the results imply that smoking history is still a significant lifestyle factor for people with diabetes and should be considered when developing diabetes prevention and treatment plans.

Research Objective 3: To examine the distribution and trends of diabetes prevalence.

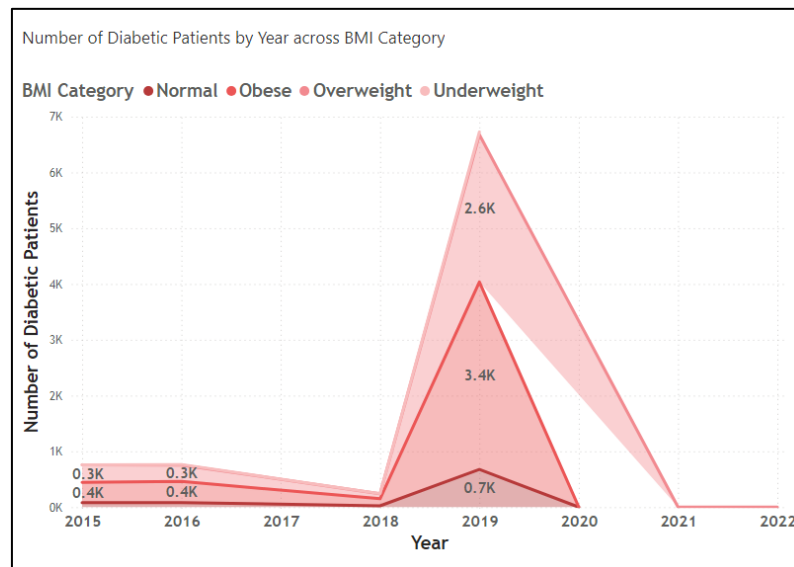
Research Question 3.1: How does the distribution of diabetes cases by gender change across different age groups?



RO2: Figure 2g Number of Diabetic Patients by Age Group Across Gender

This area chart illustrates the number of diabetic patients across different age groups, further segmented by gender. Adults represent the largest group of diabetic patients, with females representing approximately 26K cases and males 17K cases, indicating a higher number of diabetic cases among adult females compared to males. Among seniors, diabetic cases remain substantial, with around 15K females and 10K males, showing a similar gender pattern but at a lower level than adults. In contrast, younger age groups record considerably fewer diabetic cases. Children show an equal distribution between females and males at approximately 6K cases each, while teenagers record lower counts, with about 4K female and 3K male diabetic patients. Young adults reveal a moderate number of cases, with females accounting for approximately 8K cases and males 4K cases. Overall, the chart indicates that diabetic cases are most focused among adult and senior age groups, with females consistently representing a higher number of cases than males across most age categories within the dataset.

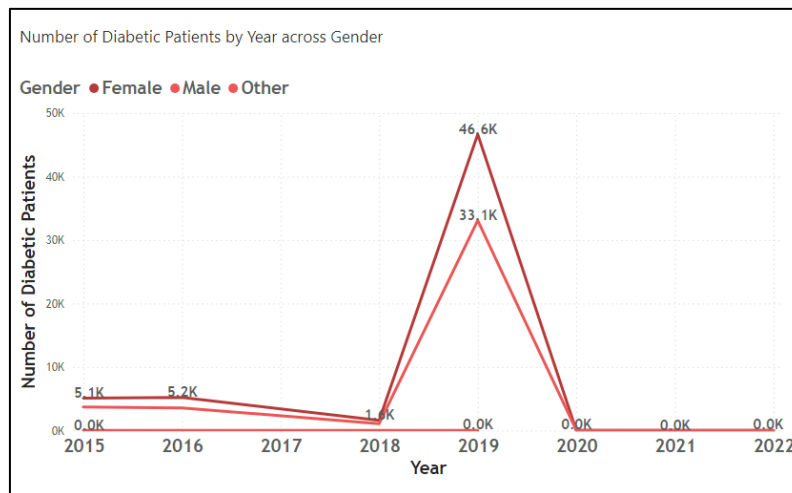
Research Question 3.2: How has the distribution of diabetes cases across BMI categories changed from 2015 to 2022?



RO2: Figure 2h Number of Diabetic Patients by Year Across BMI Category

The stacked area chart illustrates the number of diabetic patients across different BMI categories which are Normal, Obese, Overweight and Underweight from the year 2015 to 2022. From 2015 to 2018, the number of diabetic patients remains relatively low and stable across all BMI categories. A sharp increase in 2019, where the total number of diabetic patients peaks across multiple BMI groups. During this peak, the Obese category records the highest number of cases at approximately 3.4K, followed by the Overweight category at around 2.6K, while the Normal category contributes about 0.7K cases. After 2019, the number of diabetic patients declines substantially across all BMI categories, with very low values recorded in the subsequent years. Overall, the chart indicates that higher counts of diabetic patients are observed within the Obese and Overweight BMI categories during the peak period, compared to other BMI groups in the dataset.

Research Question 3.3: How has the number of diabetes cases changed from 2015 to 2022 across gender categories?



RO2: Figure 2i Number of Diabetic Patients by Year Across Gender

This line chart illustrates the trend in the number of diabetic patients across genders from 2015 to 2022. From 2015 to 2016, female diabetic cases are slightly higher than male cases, with approximately 5.1K females and 5.2K males recorded in the early years, followed by a decline toward 2018 where both genders record relatively low counts. A sharp increase is observed in 2019, where female cases peak at approximately 46.6K, while male cases reach 33.1K, representing the highest values across the period. After 2019, the number of recorded diabetic cases declines substantially, with very low or zero values observed from 2020 to 2022. Overall, the chart indicates that female patients consistently represent a higher number of recorded diabetic cases than males across the observed years, with both genders experiencing a pronounced peak in 2019 within the dataset.

5.3 CONCLUSION

This study successfully analysed diabetes patterns in the United States using a large-scale clinical dataset from 2015 to 2022, addressing all stated research objectives. The findings show that non-diabetic patients consistently outnumber diabetic patients across all years; however, diabetes prevalence is significantly higher among adults and seniors, indicating age as a key risk factor. Gender differences in diabetes prevalence are relatively small, while distribution across racial groups appears fairly uniform. Health conditions such as hypertension and heart disease are predominantly concentrated among older age groups, reinforcing the strong relationship between ageing and chronic disease burden.

Comparative analysis between diabetic and non-diabetic patients reveals clear differences in health risk indicators. Diabetic patients consistently exhibit higher average BMI, particularly within the overweight and obese categories, highlighting obesity as a major contributor to diabetes risk. In addition, diabetic individuals record substantially higher HbA1c and blood glucose levels, confirming their importance as clinical markers of diabetes. While most hypertensive and heart disease cases occur among non-diabetic patients due to population size, a stronger association between diabetes and cardiovascular conditions is observed. Smoking history analysis further suggests that former and current smokers show higher diabetes prevalence compared to non-smokers.

Finally, trend analysis indicates that diabetes prevalence varies notably across age groups and BMI categories, with adults, seniors and individuals with higher BMI consistently showing higher diabetes counts. A sharp spike in diabetes cases observed in 2019, followed by a substantial decline in subsequent years, suggests potential differences in data reporting or collection rather than a true epidemiological shift. Overall, this project demonstrates the effectiveness of interactive dashboard visualisation in transforming complex healthcare data into actionable insights, supporting data-driven decision-making for diabetes prevention, screening and public health planning.